Analysis on Finite-Dimensional Normed Vector Spaces

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Draft – Unofficial Lecture Notes

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Compactness and Attainment of Extrema

Standing conventions

Throughout, $(V, \|\cdot\|_V)$ denotes a finite-dimensional normed $\mathbb{R}$-vector space. All topological notions (open/closed/compact/continuous) are with respect to the metric induced by the norm.

Finite-dimensional spaces are not compact

Proposition 1 (A normed vector space is bounded only in the trivial case)**.** *Let $(V, \|\cdot\|_V)$ be a normed $\mathbb{R}$-vector space. Then V is bounded as a subset of itself (with respect to the metric induced by $\|\cdot\|_V$) if and only if $V = \{0\}$.*

Proof.
If $V = \{0\}$, then $\|x\|_V = 0$ for all $x \in V$, hence V is bounded.

Conversely, assume $V \neq \{0\}$ and choose $v \in V$ with $v \neq 0$. For each $t \in \mathbb{R}$, the vector tv belongs to V, and by absolute homogeneity of the norm,

$$\|tv\|_V = |t|\,\|v\|_V.$$

As $t \to \infty$, the right-hand side tends to ∞, so the set $\{\|x\|_V : x \in V\}$ is unbounded in $\mathbb{R}$. Equivalently, there is no constant $M > 0$ such that $\|x\|_V \leq M$ for all $x \in V$. Thus V is not bounded. □

Remark 1 (Compactness of a finite-dimensional normed space)**.** *Compactness is a property of* topological spaces*. Since a normed vector space $(V, \|\cdot\|_V)$ carries the topology induced by its norm, it makes sense to ask whether the underlying topological space V is compact. If V is finite-dimensional, then by Heine–Borel a subset $K \subseteq V$ is compact if and only if it is closed and bounded. Applying this with $K = V$ (which is closed in itself) shows that V is compact if and only if V is bounded as a subset of itself. By Proposition 21 this occurs only in the trivial case $V = \{0\}$.*

Compact subsets in finite dimensions

Proposition 2 (Heine–Borel in finite-dimensional normed spaces)**.** *Let $(V, \|\cdot\|_V)$ be a* finite-dimensional *normed $\mathbb{R}$-vector space, and let $K \subseteq V$. Then K is compact if and only if K is closed and bounded.*

Remark 2 (Important examples)**.** *Let $(V, \|\cdot\|_V)$ be a finite-dimensional normed $\mathbb{R}$-vector space. The* closed unit ball *and* unit sphere *are the subsets*

$$\overline{B}(0,1) := \{v \in V : \|v\|_V \leq 1\}, \qquad S := \{v \in V : \|v\|_V = 1\}.$$

Both sets are bounded. Moreover, both are closed. Hence $\overline{B}(0,1)$ and S are compact by the Heine–Borel proposition above.

Extrema on compact sets

Proposition 3 (Extreme value theorem)**.** *Let K be a compact topological space and let $f : K \to \mathbb{R}$ be continuous. Then there exist points $x_{\min}, x_{\max} \in K$ such that*

$$f(x_{\min}) = \min_{x \in K} f(x), \qquad f(x_{\max}) = \max_{x \in K} f(x).$$

Remark 3 (Continuity is essential)**.** *The conclusion of the extreme value theorem can fail if f is not continuous, even when K is compact.*

Accumulation Points and Limits at Accumulation Points

Standing conventions

Throughout this section, (X, d_X) and (Y, d_Y) denote metric spaces, and $D \subseteq X$. For $a \in X$ and $\varepsilon > 0$ we write

$$B(a, \varepsilon) := \{x \in X : \ d_X(x, a) < \varepsilon\}.$$

All limits "$x \to a$" are understood as $d_X(x, a) \to 0$ with $x \in D$, and all limits in Y are taken with respect to the metric d_Y.

Accumulation points

Definition 1 (Accumulation point (limit point) of a set)**.** *Let (X, d) be a metric space and let $D \subseteq X$. A point $a \in X$ is called an* accumulation point *of D if for every $\varepsilon > 0$ there exists $x \in D$ such that*

$$x \neq a \qquad \text{and} \qquad d(x, a) < \varepsilon.$$

Equivalently, for every $\varepsilon > 0$,

$$\big(B(a, \varepsilon) \setminus \{a\}\big) \cap D \neq \varnothing.$$

We denote the set of accumulation points of D by D'.

Remark 4 (Conceptual meaning)**.** *The statement "a is an accumulation point of D" means: no matter how closely one zooms in around a, one always finds points of D (different from a) inside that neighbourhood. In particular, a need not belong to D.*

Remark 5 (Interior point versus accumulation point)**.** *If a is an interior point of D, then there exists $\varepsilon > 0$ such that $B(a, \varepsilon) \subseteq D$. In that case a is automatically an accumulation point of D (unless $D = \{a\}$). However, an accumulation point need not be an interior point: for example, in $\mathbb{R}$ the point 0 is an accumulation point of $(0, 1)$ but is not an interior point of $(0, 1)$.*

Limits at accumulation points

Definition 2 (Limit of a function at an accumulation point)**.** *Let (X, d_X) and (Y, d_Y) be metric spaces, let $D \subseteq X$, and let $a \in X$ be an accumulation point of D. Let $f : D \to Y$ and let $q \in Y$. We say that $f(x)$* converges to q as $x \to a$ (with $x \in D$) *if for every $\varepsilon > 0$ there exists $\delta > 0$ such that*

$$x \in D, \quad 0 < d_X(x, a) < \delta \quad \Longrightarrow \quad d_Y\big(f(x), q\big) < \varepsilon.$$

In this case we write

$$\lim_{x \to a} f(x) = q.$$

Remark 6 (Conceptual meaning: what is being controlled)**.** *The limit condition compares values of $f(x)$ for points $x \in D$ that are* close to a but not equal to a. *Thus the value $f(a)$ is irrelevant, even if $a \in D$. If $a \notin D$, the definition still makes sense because only points of D near a are used.*

Remark 7 (Why a must be an accumulation point)**.** *If a is not an accumulation point of D, then for sufficiently small $\delta > 0$ there may be no points $x \in D$ with $0 < d_X(x, a) < \delta$, so the implication in the definition becomes vacuously true for every candidate q. Requiring that a is an accumulation point ensures there are always points of $D \setminus \{a\}$ arbitrarily close to a, so the limit statement has genuine content and the limit (if it exists) is unique.*

Remark 8 (Sequential characterisation (metric spaces))**.** *Let (X, d_X) and (Y, d_Y) be metric spaces. Assume a is an accumulation point of $D \subseteq X$ and let $f : D \to Y$. Then $\lim_{x\to a} f(x) = q$ if and only if for every sequence (x_n) in $D \setminus \{a\}$ with $x_n \to a$ in X, one has*

$$f(x_n) \to q \quad \text{in } Y.$$

Conceptually: no matter how *one approaches a using points of D, the values $f(x)$ must approach the same point q.*

Proposition 4 (Uniqueness of limits at an accumulation point)**.** *Let (X, d_X) and (Y, d_Y) be metric spaces, let $D \subseteq X$, and let $a \in X$ be an accumulation point of D. Let $f : D \to Y$. If*

$$\lim_{x\to a} f(x) = q \qquad \text{and} \qquad \lim_{x\to a} f(x) = q',$$

then $q = q'$.

Proof.
Assume $\lim_{x\to a} f(x) = q$ and $\lim_{x\to a} f(x) = q'$. Suppose for contradiction that $q \neq q'$. Set

$$\varepsilon := \frac{1}{2}\, d_Y(q, q') > 0.$$

By $\lim_{x\to a} f(x) = q$, there exists $\delta_1 > 0$ such that for all $x \in D$,

$$0 < d_X(x, a) < \delta_1 \quad \Longrightarrow \quad d_Y\big(f(x), q\big) < \varepsilon.$$

By $\lim_{x\to a} f(x) = q'$, there exists $\delta_2 > 0$ such that for all $x \in D$,

$$0 < d_X(x, a) < \delta_2 \quad \Longrightarrow \quad d_Y\big(f(x), q'\big) < \varepsilon.$$

Set $\delta := \min\{\delta_1, \delta_2\}$. Since a is an accumulation point of D, there exists $x \in D$ such that

$$x \neq a \qquad \text{and} \qquad d_X(x, a) < \delta.$$

For this x we have $0 < d_X(x, a) < \delta$, hence both estimates apply:

$$d_Y\big(f(x), q\big) < \varepsilon \qquad \text{and} \qquad d_Y\big(f(x), q'\big) < \varepsilon.$$

By the triangle inequality in Y,

$$d_Y(q, q') \leq d_Y\big(q, f(x)\big) + d_Y\big(f(x), q'\big) < \varepsilon + \varepsilon = 2\varepsilon = d_Y(q, q').$$

This is a contradiction. Therefore $q = q'$. $\square$

Sequential characterisation of limits

Proposition 5 (Sequential characterisation of limits in metric spaces)**.** *Let (X, d_X) and (Y, d_Y) be metric spaces, let $D \subseteq X$, and let $a \in X$ be an accumulation point of D. Let $f : D \to Y$ and let $q \in Y$. Then the following are equivalent:*

1. $\lim_{x \to a} f(x) = q$.
2. *For every sequence (x_n) with $x_n \in D \setminus \{a\}$ for all n and $x_n \to a$ in X, one has $f(x_n) \to q$ in Y.*

Remark 9 (Conceptual meaning)**.** *A sequence (x_n) with $x_n \in D \setminus \{a\}$ and $x_n \to a$ is a concrete way of* approaching a using points of D*: it is a "trial sequence" that forces points of D closer and closer to a. The sequential characterisation says that $\lim_{x \to a} f(x) = q$ exactly when* every *such approach produces the same output behaviour: no matter how one chooses points $x_n \in D \setminus \{a\}$ tending to a, the values $f(x_n)$ must tend to the single point q.*

Equivalently, there is no *sequence approaching a inside D along which $f(x_n)$ approaches a different value, or fails to converge. This is also the standard method to show that a limit does* not *exist: it suffices to find two sequences (x_n) and (y_n) in $D \setminus \{a\}$ with $x_n \to a$ and $y_n \to a$ but with $f(x_n)$ and $f(y_n)$ having different limits.*

Continuity

Standing conventions

Throughout this section, (X, d_X) and (Y, d_Y) denote metric spaces, $D \subseteq X$, and $f : D \to Y$ is a map. Fix $a \in D$. For $x \in X$ and $r > 0$ we write

$$B_X(x, r) := \{z \in X : \ d_X(z, x) < r\},$$

and for $y \in Y$ and $r > 0$ we write

$$B_Y(y, r) := \{w \in Y : \ d_Y(w, y) < r\}.$$

All limits in Y are taken with respect to the metric d_Y.

Continuity at a point

Definition 3 (Continuity at a point)**.** *Assume that a is an accumulation point of D. We say that f is* continuous at a *if*

$$\lim_{x \to a} f(x) = f(a).$$

If a is an isolated point of D, then (by convention) f is continuous at a.

Remark 10 (Conceptual meaning)**.** *Continuity at a means that f has* no jump at a*: whenever inputs $x \in D$ are close to a, the outputs $f(x)$ are close to $f(a)$. Equivalently, the value $f(a)$ agrees with the behaviour of $f(x)$ for points x near a.*

ε–δ characterisation

Proposition 6 (ε–δ characterisation of continuity)**.** *Assume that a is an accumulation point of D. Then f is continuous at a if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x \in D$,*

$$d_X(x, a) < \delta \quad \Longrightarrow \quad d_Y\big(f(x), f(a)\big) < \varepsilon.$$

Remark 11 (Conceptual meaning of the ε–δ condition)**.** *The number $\varepsilon > 0$ is an output tolerance: it prescribes how close $f(x)$ must be to $f(a)$. Continuity asserts that one can respond by choosing an input tolerance $\delta > 0$ such that every $x \in D$ within distance δ of a is sent within distance ε of $f(a)$. The key point is that, once ε is fixed, the estimate is* uniform in x near a*: the same δ must work simultaneously for all $x \in D$ with $d_X(x, a) < \delta$, and one is not allowed to choose δ depending on x.*

Sequential characterisation

Proposition 7 (Sequential characterisation of continuity in metric spaces)**.** *Assume that a is an accumulation point of D. Then f is continuous at a if and only if for every sequence (x_n) in D with $x_n \to a$ in X, one has*

$$f(x_n) \to f(a) \quad \text{in } Y.$$

Remark 12 (Conceptual meaning of the sequential characterisation)**.** *A sequence (x_n) with $x_n \to a$ is a concrete way of approaching a using points of D. The sequential characterisation says that f is continuous at a exactly when* every *such approach forces the outputs to converge to the correct value $f(a)$. Equivalently, f fails to be continuous at a precisely when there exists at least one sequence $(x_n) \subseteq D$ with $x_n \to a$ but*

$$f(x_n) \not\to f(a),$$

which exhibits a specific "bad" way of approaching a along which f does not match its value at a.

Uniform continuity

Definition 4 (Uniform continuity)**.** *Let (X, d_X) and (Y, d_Y) be metric spaces, let $D \subseteq X$, and let $f : D \to Y$. We say that f is* uniformly continuous on D *if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x, y \in D$,*

$$d_X(x, y) < \delta \quad \Longrightarrow \quad d_Y\big(f(x), f(y)\big) < \varepsilon.$$

Remark 13 (Conceptual meaning)**.** *Uniform continuity is continuity with a* uniform *input tolerance. In ordinary continuity at a point $a \in D$, the number δ in the ε–δ condition may depend on a. Uniform continuity strengthens this by requiring that, once ε is fixed, a* single *δ works simultaneously for* all *pairs of points $x, y \in D$: whenever two inputs are within δ of each other (anywhere in D), the corresponding outputs are within ε of each other. Equivalently, the function has no "regions" where one must take smaller and smaller input tolerances to achieve the same output tolerance.*

Proposition 8 (Uniform continuity implies continuity)**.** *Let (X, d_X) and (Y, d_Y) be metric spaces, let $D \subseteq X$, and let $f : D \to Y$. If f is uniformly continuous on D, then f is continuous at every point $a \in D$.*

Proof.

Assume f is uniformly continuous on D and fix $a \in D$. Let $\varepsilon > 0$ be given. By uniform continuity, there exists $\delta > 0$ such that for all $x, y \in D$,

$$d_X(x, y) < \delta \quad \Longrightarrow \quad d_Y\big(f(x), f(y)\big) < \varepsilon.$$

Now let $x \in D$ satisfy $d_X(x, a) < \delta$. Applying the implication with $y := a$ gives

$$d_Y\big(f(x), f(a)\big) < \varepsilon.$$

This is exactly the ε–δ condition for continuity of f at a. Since $a \in D$ was arbitrary, f is continuous at every point of D. □

One-Sided Limits, Divergence to $\pm\infty$, and Limits at $\pm\infty$

Standing conventions

Throughout this section, (Y, d_Y) is a metric space.

For *one-sided limits* and for *limits at* $\pm\infty$, the domain must carry a canonical order. Accordingly, in those parts we assume $D \subseteq \mathbb{R}$ and we use the usual metric $d_{\mathbb{R}}(x, y) = |x - y|$.

For *divergence to* $\pm\infty$ *at a point*, the domain may be an arbitrary metric space: in that part we assume (X, d_X) is a metric space, $D \subseteq X$, and $a \in X$ is an accumulation point of D.

One-sided limits

Definition 5 (Left-hand and right-hand limits)**.** *Let $D \subseteq \mathbb{R}$, let $a \in \mathbb{R}$, and let $f : D \to Y$. Assume that a is an accumulation point of $D \cap (-\infty, a)$ (for the left-hand limit) and an accumulation point of $D \cap (a, \infty)$ (for the right-hand limit). Let $q \in Y$.*

We say that $f(x)$ converges to q as $x \to a$ from the left*, and write*

$$\lim_{x \to a^-} f(x) = q,$$

if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x \in D$,

$$a - \delta < x < a \quad \Longrightarrow \quad d_Y\big(f(x), q\big) < \varepsilon.$$

We say that $f(x)$ converges to q as $x \to a$ from the right*, and write*

$$\lim_{x \to a^+} f(x) = q,$$

if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x \in D$,

$$a < x < a + \delta \quad \Longrightarrow \quad d_Y\big(f(x), q\big) < \varepsilon.$$

Remark 14 (Conceptual meaning)**.** *A two-sided limit $\lim_{x \to a} f(x) = q$ constrains the behaviour of $f(x)$ as x approaches a from* both *sides (within D). The left-hand limit only controls values with $x < a$ close to a, and the right-hand limit only controls values with $x > a$ close to a.*

Divergence to $\pm\infty$ at a point (general metric-space domain)

Definition 6 (Divergence to $+\infty$ and $-\infty$ as $x \to a$)**.** *Let (X, d_X) be a metric space, let $D \subseteq X$, and let $a \in X$ be an accumulation point of D. Let $f : D \to \mathbb{R}$.*

We write

$$\lim_{x \to a} f(x) = +\infty$$

if for every $M \in \mathbb{R}$ there exists $\delta > 0$ such that for all $x \in D$,

$$0 < d_X(x, a) < \delta \quad \Longrightarrow \quad f(x) > M.$$

We write

$$\lim_{x \to a} f(x) = -\infty$$

if for every $M \in \mathbb{R}$ there exists $\delta > 0$ such that for all $x \in D$,

$$0 < d_X(x, a) < \delta \quad \Longrightarrow \quad f(x) < M.$$

Remark 15 (Conceptual meaning)**.** *The domain being a general metric space only changes what it means to approach a: it means $d_X(x, a) \to 0$ with $x \in D$. Divergence to $+\infty$ means that, as one forces x into smaller and smaller balls around a (excluding a), the real values $f(x)$ are forced above every prescribed threshold M; divergence to $-\infty$ is the analogous statement with $f(x)$ forced below every threshold.*

Limits at $+\infty$ and $-\infty$

Definition 7 (Limit as $x \to +\infty$ and as $x \to -\infty$)**.** *Let $D \subseteq \mathbb{R}$ and let $f : D \to Y$. Let $q \in Y$.*

We write

$$\lim_{x \to +\infty} f(x) = q$$

if for every $\varepsilon > 0$ there exists $R > 0$ such that for all $x \in D$,

$$x > R \quad \Longrightarrow \quad d_Y\big(f(x), q\big) < \varepsilon.$$

We write

$$\lim_{x \to -\infty} f(x) = q$$

if for every $\varepsilon > 0$ there exists $R > 0$ such that for all $x \in D$,

$$x < -R \quad \Longrightarrow \quad d_Y\big(f(x), q\big) < \varepsilon.$$

Remark 16 (Conceptual meaning)**.** *A limit at $+\infty$ is a statement about the* right tail *of the domain: far enough to the right (that is, for all $x > R$), all values $f(x)$ lie within a prescribed tolerance of q. A limit at $-\infty$ is the analogous statement about the* left tail *(for all $x < -R$). In particular, the values of f on any bounded region are irrelevant to limits at $\pm\infty$.*

Coordinate Functions of a Map

Standing conventions

Throughout, $\mathbb{F}$ denotes either $\mathbb{R}$ or $\mathbb{C}$. A *normed $\mathbb{F}$-vector space* is a pair $(W, \|\cdot\|_W)$ where W is a vector space over $\mathbb{F}$ and $\|\cdot\|_W : W \to [0,\infty)$ satisfies, for all $x, y \in W$ and $\lambda \in \mathbb{F}$,

$$\|x\|_W = 0 \iff x = 0, \qquad \|\lambda x\|_W = |\lambda|\,\|x\|_W, \qquad \|x+y\|_W \le \|x\|_W + \|y\|_W,$$

where $|\cdot|$ is the usual absolute value on $\mathbb{R}$ or the modulus on $\mathbb{C}$.

In this note we assume all vector spaces are *finite-dimensional* over $\mathbb{F}$, so that bases and coordinate maps are available in the usual sense.

Coordinates in a finite-dimensional space

Definition 8 (Coordinate functionals associated to a basis)**.** *Let $(W, \|\cdot\|_W)$ be a finite-dimensional normed $\mathbb{F}$-vector space and let*

$$\beta = (w_1, \dots, w_m)$$

be an ordered basis of W. For each $j \in \{1, \dots, m\}$ define the map $\pi_j^\beta : W \to \mathbb{F}$ by the rule: for $y \in W$, write y uniquely as

$$y = \sum_{k=1}^{m} y_k w_k \qquad (y_k \in \mathbb{F}),$$

and set

$$\pi_j^\beta(y) := y_j.$$

The maps $\pi_1^\beta, \dots, \pi_m^\beta$ are called the coordinate functionals *of W with respect to β.*

Proposition 9 (Basic properties of coordinate functionals)**.** *With notation as above:*

1. *Each $\pi_j^\beta : W \to \mathbb{F}$ is $\mathbb{F}$-linear.*
2. *Every $y \in W$ satisfies the reconstruction formula*

$$y = \sum_{j=1}^{m} \pi_j^\beta(y)\, w_j.$$

Proof.
(1) Let $y, z \in W$ and $\lambda \in \mathbb{F}$. Write $y = \sum_{k=1}^m y_k w_k$ and $z = \sum_{k=1}^m z_k w_k$. Then $y+z = \sum_{k=1}^m (y_k + z_k) w_k$ and $\lambda y = \sum_{k=1}^m (\lambda y_k) w_k$ because vector space addition and scalar multiplication distribute over finite sums: one uses

$$(\alpha + \beta) w = \alpha w + \beta w$$

to combine the k-th terms in $y + z$, and one uses

$$\lambda(u+v) = \lambda u + \lambda v \quad \text{together with} \quad \lambda(\alpha w) = (\lambda\alpha) w$$

to distribute λ over the finite linear combination. By definition of π_j^β this gives

$$\pi_j^\beta(y+z) = y_j + z_j = \pi_j^\beta(y) + \pi_j^\beta(z), \qquad \pi_j^\beta(\lambda y) = \lambda y_j = \lambda\, \pi_j^\beta(y),$$

so π_j^β is linear.

(2) If $y = \sum_{k=1}^m y_k w_k$, then $\pi_j^\beta(y) = y_j$ for each j, hence $\sum_{j=1}^m \pi_j^\beta(y)\, w_j = \sum_{j=1}^m y_j w_j = y$. $\square$

Remark 17 (Continuity)**.** *Since W is finite-dimensional, every linear map from W to $\mathbb{F}$ is continuous (equivalently, bounded). In particular, each coordinate functional π_j^β is continuous.*

Component functions of a map

Definition 9 (Component (coordinate) functions of a map)**.** *Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed $\mathbb{F}$-vector spaces, with W finite-dimensional, and let $f : V \to W$ be a map. Fix an ordered basis $\beta = (w_1, \ldots, w_m)$ of W and let $\pi_1^\beta, \ldots, \pi_m^\beta : W \to \mathbb{F}$ be the associated coordinate functionals. For each $j \in \{1, \ldots, m\}$ define the j-th* component function of f with respect to β *by*

$$f_j := \pi_j^\beta \circ f : V \to \mathbb{F}.$$

Proposition 10 (Reconstruction of f from its component functions)**.** *With the notation above, for every $x \in V$,*

$$f(x) = \sum_{j=1}^{m} f_j(x)\, w_j.$$

Proof. Fix $x \in V$ and set $y := f(x) \in W$. By the reconstruction formula in W,

$$y = \sum_{j=1}^{m} \pi_j^\beta(y)\, w_j.$$

Substituting $y = f(x)$ and using $f_j = \pi_j^\beta \circ f$ gives

$$f(x) = \sum_{j=1}^{m} \pi_j^\beta(f(x))\, w_j = \sum_{j=1}^{m} f_j(x)\, w_j.$$

□

Remark 18 (Real vs complex scalars)**.** *If $\mathbb{F} = \mathbb{R}$ then each component function satisfies $f_j : V \to \mathbb{R}$. If $\mathbb{F} = \mathbb{C}$ then each component function satisfies $f_j : V \to \mathbb{C}$. In general, the component functions take values in the base field $\mathbb{F}$, not necessarily in $\mathbb{R}$.*

Example: Reconstruction in W and Reconstruction of f

Let $\mathbb{F} = \mathbb{R}$. Let $V = \mathbb{R}^2$ and $W = \mathbb{R}^2$ with the Euclidean norm. Fix the ordered basis

$$\beta = (w_1, w_2) \quad \text{of } W, \qquad w_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad w_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

For $y \in W$, write uniquely

$$y = y_1 w_1 + y_2 w_2,$$

and define the coordinate functionals $\pi_1^\beta, \pi_2^\beta : W \to \mathbb{R}$ by

$$\pi_1^\beta(y) = y_1, \qquad \pi_2^\beta(y) = y_2.$$

Example 1 (Reconstruction of a vector $y \in W$)**.** *Let*

$$y = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \in W.$$

We find its coordinates with respect to β by solving

$$\begin{bmatrix} 3 \\ 1 \end{bmatrix} = y_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + y_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} \implies \begin{bmatrix} 1 & 1 & 3 \\ 1 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix}$$

Thus the system

$$y_1 + y_2 = 3, \qquad y_1 - y_2 = 1$$

gives $y_1 = 2$ *and* $y_2 = 1$. *Hence*

$$\pi_1^\beta(y) = 2, \qquad \pi_2^\beta(y) = 1,$$

and the reconstruction formula in W *reads*

$$y = \pi_1^\beta(y)\, w_1 + \pi_2^\beta(y)\, w_2 = 2w_1 + 1w_2 = 2\begin{bmatrix}1\\1\end{bmatrix} + \begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}3\\1\end{bmatrix}.$$

Example 2 (Reconstruction of a map $f : V \to W$ from component functions)**.** *Define a map* $f : V \to W$ *by*

$$f(x_1, x_2) = \begin{bmatrix}x_1 + x_2\\x_1 - x_2\end{bmatrix}.$$

For each $j \in \{1, 2\}$ *define the component function with respect to* β

$$f_j := \pi_j^\beta \circ f : V \to \mathbb{R}.$$

To compute f_1, f_2 *explicitly, fix* $(x_1, x_2) \in V$ *and write* $f(x_1, x_2) = a_1 w_1 + a_2 w_2$. *That is, solve*

$$\begin{bmatrix}x_1 + x_2\\x_1 - x_2\end{bmatrix} = a_1\begin{bmatrix}1\\1\end{bmatrix} + a_2\begin{bmatrix}1\\-1\end{bmatrix} \implies \begin{bmatrix}1 & 1 & x_1 + x_2\\1 & -1 & x_1 - x_2\end{bmatrix} \sim \begin{bmatrix}1 & 0 & x_2\\0 & 1 & x_2\end{bmatrix}$$

Thus the system

$$a_1 + a_2 = x_1 + x_2, \qquad a_1 - a_2 = x_1 - x_2,$$

gives $a_1 = x_1$ *and* $a_2 = x_2$. *Therefore,*

$$f_1(x_1, x_2) = \pi_1^\beta(f(x_1, x_2)) = a_1 = x_1, \qquad f_2(x_1, x_2) = \pi_2^\beta(f(x_1, x_2)) = a_2 = x_2.$$

The reconstruction formula for f *with respect to* β *is then, for all* $(x_1, x_2) \in V$,

$$f(x_1, x_2) = f_1(x_1, x_2)\, w_1 + f_2(x_1, x_2)\, w_2 = x_1\begin{bmatrix}1\\1\end{bmatrix} + x_2\begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}x_1 + x_2\\x_1 - x_2\end{bmatrix}.$$

The Fréchet Derivative

Standing conventions

Throughout, $\mathbb{F}$ denotes $\mathbb{R}$. A *normed* $\mathbb{F}$*-vector space* is a pair $(W, \|\cdot\|_W)$ where W is a vector space over $\mathbb{F}$ and $\|\cdot\|_W : W \to [0, \infty)$ satisfies, for all $x, y \in W$ and $\lambda \in \mathbb{F}$,

$$\|x\|_W = 0 \iff x = 0, \qquad \|\lambda x\|_W = |\lambda|\, \|x\|_W, \qquad \|x + y\|_W \le \|x\|_W + \|y\|_W,$$

where $|\cdot|$ is the usual absolute value on $\mathbb{R}$.

All limits are taken with respect to the metrics induced by the norms. In particular, for a map $f : \Omega \to W$ and a point $a \in \Omega$, the expression $x \to a$ means $\|x - a\|_V \to 0$.

In this note, $\Omega \subseteq V$ will always denote an *open* subset of a normed space $(V, \|\cdot\|_V)$.

Linear approximation and remainder

Definition 10 (Remainder associated to a candidate linear approximation)**.** *Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed $\mathbb{R}$-vector spaces, let $\Omega \subseteq V$ be open, let $f : \Omega \to W$, and let $a \in \Omega$. Let $A : V \to W$ be a linear map. Define the* remainder (error) of f at a relative to A *to be the map*

$$r : \{h \in V : a + h \in \Omega\} \to W, \qquad r(h) := f(a + h) - f(a) - A(h).$$

Equivalently, for $x \in \Omega$ one may write

$$\mathrm{Err}(x) := f(x) - f(a) - A(x - a),$$

and then $\mathrm{Err}(a + h) = r(h)$ whenever $a + h \in \Omega$.

Remark 19 (The identity $f(a + h) = f(a) + A(h) + r(h)$ is definitional)**.** *For every h with $a + h \in \Omega$, one has*

$$f(a + h) = f(a) + A(h) + r(h).$$

This is not an additional assumption: it is simply a rewriting of the definition of $r(h)$.

Fréchet differentiability

Definition 11 (Fréchet differentiability at a point)**.** *Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed $\mathbb{R}$-vector spaces, let $\Omega \subseteq V$ be open, let $f : \Omega \to W$, and let $a \in \Omega$. We say that f is* Fréchet differentiable at a *if there exists a bounded linear map $A : V \to W$ such that*

$$\lim_{h \to 0} \frac{\|r(h)\|_W}{\|h\|_V} = 0,$$

that is,

$$\lim_{h \to 0} \frac{\|f(a + h) - f(a) - A(h)\|_W}{\|h\|_V} = 0.$$

In that case, the map A is unique and is denoted by $(Df)_a$; it is called the Fréchet derivative of f at a.

Remark 20 (ε–δ form of the definition)**.** *Let $f : \Omega \to W$ and $a \in \Omega$. The condition*

$$\lim_{h \to 0} \frac{\|f(a + h) - f(a) - A(h)\|_W}{\|h\|_V} = 0$$

is equivalent to the following statement: for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $h \in V$,

$$0 < \|h\|_V < \delta \quad \text{and} \quad a + h \in \Omega \quad \Longrightarrow \quad \|f(a + h) - f(a) - A(h)\|_W \le \varepsilon \, \|h\|_V.$$

In particular, the definition asserts that the remainder is bounded by an arbitrarily small multiple of $\|h\|_V$ for all sufficiently small h.

Linear maps are Fréchet differentiable

Proposition 11 (Linear maps are Fréchet differentiable)**.** *Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed $\mathbb{R}$-vector spaces, and let $L \in B(V, W)$. Then L is Fréchet differentiable at every $a \in V$, and*

$$(DL)_a = L \qquad (a \in V).$$

Proof.
Fix $a \in V$. Consider the bounded linear map $A : V \to W$ defined by $A := L$. For $h \in V$,

$$L(a+h) - L(a) - A(h) = L(a+h) - L(a) - L(h) = 0,$$

using linearity of L. Therefore, for every $h \neq 0$,

$$\frac{\|L(a+h) - L(a) - A(h)\|_W}{\|h\|_V} = \frac{\|0\|_W}{\|h\|_V} = 0,$$

so the limit as $h \to 0$ is 0. Hence L is Fréchet differentiable at a with $(DL)_a = A = L$. □

Proposition 12 (Why the denominator matters)**.** *Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed $\mathbb{R}$-vector spaces, let $\Omega \subseteq V$ be open, let $f : \Omega \to W$, and let $a \in \Omega$. Fix a linear map $A : V \to W$ and define $r(h) = f(a+h) - f(a) - A(h)$. The implication*

$$\left(\|r(h)\|_W \to 0 \text{ as } h \to 0 \right) \implies \left(\frac{\|r(h)\|_W}{\|h\|_V} \to 0 \text{ as } h \to 0 \right)$$

is false in general.

Proof.
Take $V = W = \mathbb{R}$ with the usual absolute value as the norm, and define a function $r : \mathbb{R} \to \mathbb{R}$ by $r(h) = |h|$. Then $r(h) \to 0$ as $h \to 0$, but for every $h \neq 0$,

$$\frac{|r(h)|}{|h|} = \frac{|h|}{|h|} = 1,$$

so the quotient does not tend to 0. □

Remark 21 (Conceptual summary)**.** *Let $f : \Omega \to W$ and $a \in \Omega$. Fréchet differentiability at a asserts the existence of a bounded linear map $(Df)_a : V \to W$ such that, for small increments h,*

$$f(a+h) = f(a) + (Df)_a(h) + r(h), \qquad r(h) := f(a+h) - f(a) - (Df)_a(h),$$

with the quantitative control

$$\frac{\|r(h)\|_W}{\|h\|_V} \to 0 \qquad (h \to 0).$$

Thus $f(a) + (Df)_a(h)$ provides the correct first-order (linear) approximation *of f near a, and the remaining term is negligible compared with $\|h\|_V$.*

Example: A Simple Computation in $\mathbb{R}^2$

Example 3 (A function with derivative 0 at the origin)**.** *Let $V = \mathbb{R}^2$ and $W = \mathbb{R}$ with the Euclidean norm on $\mathbb{R}^2$ and the absolute value on $\mathbb{R}$. Define $f : \mathbb{R}^2 \to \mathbb{R}$ by*

$$f(x_1, x_2) = x_1^2 + x_2^2,$$

and let $a = (0, 0)$. We claim that f is Fréchet differentiable at a and that $(Df)_a$ is the zero linear map.

Let $A : \mathbb{R}^2 \to \mathbb{R}$ be the zero map, so $A(h) = 0$ for all $h \in \mathbb{R}^2$. For $h = (h_1, h_2)$ we have $a + h = h$ and $f(a) = 0$, so

$$f(a + h) - f(a) - A(h) = f(h_1, h_2) = h_1^2 + h_2^2.$$

Hence, for $h \neq 0$,

$$\frac{|f(a + h) - f(a) - A(h)|}{\|h\|_2} = \frac{h_1^2 + h_2^2}{\sqrt{h_1^2 + h_2^2}} = \sqrt{h_1^2 + h_2^2} = \|h\|_2 \longrightarrow 0 \quad (h \to 0).$$

Therefore f is Fréchet differentiable at a and $(Df)_a = A$, i.e. $(Df)_{(0,0)}$ is the zero linear map.

The One-Variable Derivative

Standing conventions

Throughout, $\mathbb{F}$ denotes $\mathbb{R}$. A *normed $\mathbb{F}$-vector space* is a pair $(W, \|\cdot\|_W)$ where W is a vector space over $\mathbb{F}$ and $\|\cdot\|_W : W \to [0, \infty)$ satisfies, for all $x, y \in W$ and $\lambda \in \mathbb{F}$,

$$\|x\|_W = 0 \iff x = 0, \qquad \|\lambda x\|_W = |\lambda|\, \|x\|_W, \qquad \|x + y\|_W \leq \|x\|_W + \|y\|_W,$$

where $|\cdot|$ is the usual absolute value on $\mathbb{R}$.

All limits in W are taken with respect to the norm $\|\cdot\|_W$. In particular, for a function $q : J \to W$ and a point $t_0 \in J$, the statement

$$\lim_{t \to t_0} q(t) = w$$

means

$$\lim_{t \to t_0} \|q(t) - w\|_W = 0.$$

The one-variable derivative

Definition 12 (Derivative at a point)**.** *Let $I \subseteq \mathbb{R}$ be an open interval, let $(W, \|\cdot\|_W)$ be a normed $\mathbb{R}$-vector space, and let $g : I \to W$ be a map. Fix $t_0 \in I$. If there exists $w \in W$ such that*

$$\lim_{h \to 0} \left\| \frac{g(t_0 + h) - g(t_0)}{h} - w \right\|_W = 0,$$

then we say that g is differentiable at t_0 *and we define the* derivative of g at t_0 *by*

$$g'(t_0) := w.$$

Remark 22 (Equivalent limit notation)**.** *The condition*

$$\lim_{h\to 0}\left\|\frac{g(t_0+h)-g(t_0)}{h}-w\right\|_W=0$$

is equivalent to the statement

$$\lim_{h\to 0}\frac{g(t_0+h)-g(t_0)}{h}=w,$$

where the latter limit is understood as a limit in the normed space W.

Proposition 13 (First-order approximation in one variable)**.** *Let $I\subseteq\mathbb{R}$ be open, let $g:I\to W$, and let $t_0\in I$. Then g is differentiable at t_0 with $g'(t_0)=w$ if and only if there exists a map*

$$r:\{h\in\mathbb{R}:t_0+h\in I\}\to W$$

such that for all h with $t_0+h\in I$,

$$g(t_0+h)=g(t_0)+h\,w+r(h),$$

and

$$\lim_{h\to 0}\frac{\|r(h)\|_W}{|h|}=0.$$

Proof.
Assume g is differentiable at t_0 with derivative w. Define $r(h):=g(t_0+h)-g(t_0)-h\,w$. Then the identity

$$g(t_0+h)=g(t_0)+h\,w+r(h)$$

holds by definition of $r(h)$. For $h\neq 0$,

$$\frac{\|r(h)\|_W}{|h|}=\left\|\frac{g(t_0+h)-g(t_0)-h\,w}{h}\right\|_W=\left\|\frac{g(t_0+h)-g(t_0)}{h}-w\right\|_W,$$

which tends to 0 by the definition of differentiability. Hence $\|r(h)\|_W/|h|\to 0$.

Conversely, assume there is a map r such that $g(t_0+h)=g(t_0)+h\,w+r(h)$ and $\|r(h)\|_W/|h|\to 0$. Then for $h\neq 0$,

$$\frac{g(t_0+h)-g(t_0)}{h}-w=\frac{r(h)}{h},$$

so

$$\left\|\frac{g(t_0+h)-g(t_0)}{h}-w\right\|_W=\frac{\|r(h)\|_W}{|h|}\longrightarrow 0.$$

Thus g is differentiable at t_0 and $g'(t_0)=w$. □

Fréchet differentiability on $\mathbb{R}$

Definition 13 (Fréchet differentiability for maps $I\subseteq\mathbb{R}\to W$)**.** *Let $I\subseteq\mathbb{R}$ be open and let $g:I\to W$. Fix $t_0\in I$. We say that g is* Fréchet differentiable at t_0 *if there exists a bounded linear map $A:\mathbb{R}\to W$ such that*

$$\lim_{h\to 0}\frac{\|g(t_0+h)-g(t_0)-A(h)\|_W}{|h|}=0.$$

In that case A is unique and is denoted by $(Dg)_{t_0}$.

Proposition 14 (Linear maps $\mathbb{R} \to W$)**.** *Let $A : \mathbb{R} \to W$ be linear. Then for all $h \in \mathbb{R}$,*

$$A(h) = h\,A(1).$$

In particular, every linear map $A : \mathbb{R} \to W$ is determined by the single vector $A(1) \in W$.

Proof.
For $h \in \mathbb{R}$ one has $h = h \cdot 1$, hence by linearity,

$$A(h) = A(h \cdot 1) = h\,A(1).$$

□

Connection to Fréchet differentiability

Proposition 15 (Equivalence in one variable)**.** *Let $I \subseteq \mathbb{R}$ be open, let $g : I \to W$, and let $t_0 \in I$. Then the following are equivalent:*

1. *g is differentiable at t_0 in the one-variable sense, and $g'(t_0)$ exists in W.*
2. *g is Fréchet differentiable at t_0 as a map $I \subseteq \mathbb{R} \to W$.*

Moreover, if these conditions hold, then the Fréchet derivative $(Dg)_{t_0} : \mathbb{R} \to W$ is the linear map

$$(Dg)_{t_0}(h) = h\,g'(t_0) \qquad (h \in \mathbb{R}),$$

and conversely,

$$g'(t_0) = (Dg)_{t_0}(1).$$

Proof.
(1)⇒(2). Assume g is differentiable at t_0 and set $w := g'(t_0)$. Define a linear map $A : \mathbb{R} \to W$ by $A(h) := h\,w$. By Proposition 4, the differentiability of g at t_0 is equivalent to the existence of a remainder $r(h)$ such that

$$g(t_0 + h) = g(t_0) + h\,w + r(h) \quad \text{and} \quad \frac{\|r(h)\|_W}{|h|} \to 0.$$

But then $g(t_0 + h) - g(t_0) - A(h) = r(h)$, hence

$$\frac{\|g(t_0 + h) - g(t_0) - A(h)\|_W}{|h|} = \frac{\|r(h)\|_W}{|h|} \to 0.$$

Thus g is Fréchet differentiable at t_0 with $(Dg)_{t_0} = A$, i.e. $(Dg)_{t_0}(h) = h\,g'(t_0)$.

(2)⇒(1). Assume g is Fréchet differentiable at t_0. Then there exists a linear map $A : \mathbb{R} \to W$ such that

$$\frac{\|g(t_0 + h) - g(t_0) - A(h)\|_W}{|h|} \to 0.$$

By Proposition 5, $A(h) = h\,A(1)$. Set $w := A(1)$, so $A(h) = h\,w$. Then

$$\frac{\|g(t_0 + h) - g(t_0) - h\,w\|_W}{|h|} \to 0,$$

which is equivalent to

$$\left\| \frac{g(t_0 + h) - g(t_0)}{h} - w \right\|_W \to 0.$$

Hence g is differentiable at t_0 and $g'(t_0) = w = A(1) = (Dg)_{t_0}(1)$. □

Remark 23 (Interpretation)**.** *In one variable, the usual derivative $g'(t_0) \in W$ and the Fréchet derivative $(Dg)_{t_0} : \mathbb{R} \to W$ carry the same information. The scalar increment $h \in \mathbb{R}$ is converted into an output increment in W by multiplication with the vector $g'(t_0)$, namely*

$$(Dg)_{t_0}(h) = h\, g'(t_0).$$

Example: A Computation in $W = \mathbb{R}^2$

Example 4 (A linear example)**.** *Let $W = \mathbb{R}^2$ with the Euclidean norm and let $I = \mathbb{R}$. Define $g : \mathbb{R} \to \mathbb{R}^2$ by*

$$g(t) = \begin{bmatrix} t \\ 2t \end{bmatrix}.$$

Fix $t_0 \in \mathbb{R}$. For $h \neq 0$,

$$\frac{g(t_0 + h) - g(t_0)}{h} = \frac{1}{h}\left(\begin{bmatrix} t_0 + h \\ 2(t_0 + h) \end{bmatrix} - \begin{bmatrix} t_0 \\ 2t_0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Thus the limit exists and

$$g'(t_0) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

By Proposition 6, the Fréchet derivative $(Dg)_{t_0} : \mathbb{R} \to \mathbb{R}^2$ is given by

$$(Dg)_{t_0}(h) = h\, g'(t_0) = h \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} h \\ 2h \end{bmatrix}.$$

Directional Derivatives

Standing conventions

Throughout, $\mathbb{F}$ denotes $\mathbb{R}$. A *normed $\mathbb{F}$-vector space* is a pair $(W, \|\cdot\|_W)$ where W is a vector space over $\mathbb{F}$ and $\|\cdot\|_W : W \to [0, \infty)$ satisfies, for all $x, y \in W$ and $\lambda \in \mathbb{F}$,

$$\|x\|_W = 0 \iff x = 0, \qquad \|\lambda x\|_W = |\lambda|\, \|x\|_W, \qquad \|x + y\|_W \leq \|x\|_W + \|y\|_W,$$

where $|\cdot|$ is the usual absolute value on $\mathbb{R}$.

Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed $\mathbb{R}$-vector spaces. Let $\Omega \subseteq V$ be open, and let $f : \Omega \to W$ be a map. Fix $a \in \Omega$.

All limits in W are taken with respect to $\|\cdot\|_W$. In particular, for a function $q : J \to W$ and a point $t_0 \in J$, the statement

$$\lim_{t \to t_0} q(t) = w$$

means

$$\lim_{t \to t_0} \|q(t) - w\|_W = 0.$$

Restricting f to a line through a

Proposition 16 (Local existence of a line segment in Ω)**.** *Let $v \in V$. Since Ω is open and $a \in \Omega$, there exists $r > 0$ such that $B(a, r) \subseteq \Omega$. If $v \neq 0$, set*

$$I := \left(-\frac{r}{\|v\|_V}, \frac{r}{\|v\|_V}\right).$$

If $v = 0$, set $I := (-1, 1)$. Then I is an open interval containing 0 and

$$a + tv \in \Omega \qquad \text{for all } t \in I.$$

Proof.
Choose $r > 0$ such that $B(a, r) \subseteq \Omega$.

If $v \neq 0$ and $t \in I$, then $|t| \, \|v\|_V < r$. Hence

$$\|a + tv - a\|_V = \|tv\|_V = |t| \, \|v\|_V < r,$$

so $a + tv \in B(a, r) \subseteq \Omega$.

If $v = 0$ then $a + tv = a \in \Omega$ for all t, so any open interval containing 0 works; in particular $(-1, 1)$ does. □

Definition 14 (Restriction of f to the line through a in direction v)**.** *Let $v \in V$ and let I be an open interval containing 0 such that $a + tv \in \Omega$ for all $t \in I$ (for instance, I as in Proposition 8). Define*

$$g : I \to W, \qquad g(t) := f(a + tv).$$

Directional derivatives

Definition 15 (Directional derivative)**.** *Let $v \in V$ and let $g : I \to W$ be defined by $g(t) = f(a+tv)$ on an interval I as above. If g is differentiable at 0, we define the* directional derivative of f at a in direction v *to be*

$$(D_v f)_a := g'(0).$$

Equivalently, if the limit exists in W, one has

$$(D_v f)_a = \lim_{t \to 0} \frac{f(a + tv) - f(a)}{t}.$$

Remark 24 (First-order approximation along the line)**.** *If $(D_v f)_a$ exists, then there is a remainder $\rho(t) \in W$ such that for all t sufficiently close to 0,*

$$f(a + tv) = f(a) + t \, (D_v f)_a + \rho(t),$$

and

$$\lim_{t \to 0} \frac{\|\rho(t)\|_W}{|t|} = 0.$$

Thus the directional derivative captures the first-order behaviour of f along the straight line $t \mapsto a + tv$.

Proposition 17 (Two basic properties)**.** *Assume $(D_v f)_a$ exists.*

1. *If $v = 0$, then $(D_0 f)_a = 0$.*

2. For every $\lambda \in \mathbb{R}$, one has $(D_{\lambda v} f)_a = \lambda\,(D_v f)_a$.

Proof.
(1) If $v = 0$ then $f(a+tv) = f(a)$ for all t, so the difference quotient is 0 for all $t \neq 0$ and hence the limit is 0.

(2) Let $\lambda \in \mathbb{R}$. If $\lambda = 0$ this is part (1). Assume $\lambda \neq 0$. For $t \neq 0$ set $s = \lambda t$, so that $s \to 0$ as $t \to 0$. Then

$$\frac{f(a + t(\lambda v)) - f(a)}{t} = \frac{f(a + sv) - f(a)}{s/\lambda} = \lambda \frac{f(a + sv) - f(a)}{s}.$$

Taking limits as $t \to 0$ (equivalently $s \to 0$) gives $(D_{\lambda v} f)_a = \lambda (D_v f)_a$. □

Connection to Fréchet differentiability

Proposition 18 (Fréchet differentiability implies existence of all directional derivatives)**.** *Assume f is Fréchet differentiable at a, with Fréchet derivative $(Df)_a : V \to W$. Then for every $v \in V$, the directional derivative $(D_v f)_a$ exists and satisfies*

$$(D_v f)_a = (Df)_a(v).$$

Proof.
Since f is Fréchet differentiable at a, there exists a remainder $r : \{h \in V : a + h \in \Omega\} \to W$ such that

$$f(a + h) = f(a) + (Df)_a(h) + r(h), \qquad \lim_{h \to 0} \frac{\|r(h)\|_W}{\|h\|_V} = 0.$$

Fix $v \in V$. For t sufficiently close to 0 we have $a + tv \in \Omega$ and thus

$$f(a + tv) - f(a) = (Df)_a(tv) + r(tv) = t\,(Df)_a(v) + r(tv).$$

For $t \neq 0$,

$$\frac{f(a + tv) - f(a)}{t} = (Df)_a(v) + \frac{r(tv)}{t}.$$

Moreover,

$$\left\| \frac{r(tv)}{t} \right\|_W = \frac{\|r(tv)\|_W}{|t|} = \frac{\|r(tv)\|_W}{\|tv\|_V}\,\|v\|_V \longrightarrow 0 \cdot \|v\|_V = 0,$$

because $\|tv\|_V = |t|\,\|v\|_V$ and $\|r(h)\|_W / \|h\|_V \to 0$ as $h \to 0$. Hence the difference quotient converges to $(Df)_a(v)$, so $(D_v f)_a$ exists and equals $(Df)_a(v)$. □

Remark 25 (Why the converse can fail)**.** *Directional derivatives are defined by restricting f to one line at a time. Consequently, even if $(D_v f)_a$ exists for every $v \in V$, f may fail to be Fréchet differentiable at a. Fréchet differentiability requires a single bounded linear map $A : V \to W$ such that*

$$\frac{\|f(a + h) - f(a) - A(h)\|_W}{\|h\|_V} \to 0 \qquad (h \to 0),$$

which is a uniform first-order approximation (valid for all small increments h simultaneously). Equivalently, the condition above is the ε–δ statement: for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $h \in V$,

$$0 < \|h\|_V < \delta \;\text{ and }\; a + h \in \Omega \quad \Longrightarrow \quad \|f(a + h) - f(a) - A(h)\|_W \leq \varepsilon\,\|h\|_V.$$

In this sense, once ε is fixed, the same δ works simultaneously for every direction and every vector h with $\|h\|_V < \delta$; there is no choice of δ that depends on the direction. Directional differentiability provides only line-by-line information and need not assemble into such a single linear approximation.

Example: Directional derivatives exist but no Fréchet derivative

Example 5 (A standard counterexample in $\mathbb{R}^2$)**.** *Let $V = \mathbb{R}^2$ with the Euclidean norm and let $W = \mathbb{R}$ with the absolute value. Define $f : \mathbb{R}^2 \to \mathbb{R}$ by*

$$f(x_1, x_2) = \begin{cases} \dfrac{x_1^2 x_2}{x_1^4 + x_2^2}, & (x_1, x_2) \neq (0,0), \\ 0, & (x_1, x_2) = (0,0). \end{cases}$$

Let $a = (0,0)$.
Directional derivatives at a. *Fix $v = (v_1, v_2) \in \mathbb{R}^2$. For $t \neq 0$,*

$$\frac{f(tv_1, tv_2) - f(0,0)}{t} = \frac{1}{t} \cdot \frac{t^3 v_1^2 v_2}{t^4 v_1^4 + t^2 v_2^2} = \frac{v_1^2 v_2}{t^2 v_1^4 + v_2^2}.$$

If $v_2 \neq 0$ then $t^2 v_1^4 + v_2^2 \to v_2^2$ as $t \to 0$, hence

$$\lim_{t \to 0} \frac{f(tv_1, tv_2) - f(0,0)}{t} = \frac{v_1^2 v_2}{v_2^2} = \frac{v_1^2}{v_2}.$$

If $v_2 = 0$ then $f(tv_1, 0) = 0$ for all t, hence the difference quotient is 0 and the limit is 0. Thus $(D_v f)_{(0,0)}$ exists for every $v \in \mathbb{R}^2$.

Failure of Fréchet differentiability at a. *Assume for contradiction that f is Fréchet differentiable at a with derivative $A : \mathbb{R}^2 \to \mathbb{R}$. By Proposition 9, for every v one has $(D_v f)_a = A(v)$, so $v \mapsto (D_v f)_a$ must be linear. However, using the values computed above,*

$$(D_{(1,1)} f)_{(0,0)} = 1, \qquad (D_{(1,0)} f)_{(0,0)} = 0 \qquad (D_{(2,1)} f)_{(0,0)} = 4$$

Linearity would therefore imply

$$(D_{(2,1)} f)_{(0,0)} = A(2,1) = A(1,1) + A(1,0) = (D_{(1,1)} f)_{(0,0)} + (D_{(1,0)} f)_{(0,0)} = 1 + 0 = 1,$$

contradicting $(D_{(2,1)} f)_{(0,0)} = 4$. Hence no such linear map A exists and f is not Fréchet differentiable at a.

Partial Derivatives

Standing conventions

Throughout, $\mathbb{F}$ denotes $\mathbb{R}$. Let $\Omega \subseteq \mathbb{R}^d$ be open and let $(W, \|\cdot\|_W)$ be a normed $\mathbb{R}$-vector space. Let $f : \Omega \to W$ be a map and fix $a = (a_1, \ldots, a_d) \in \Omega$.

For $i \in \{1, \ldots, d\}$, let $e_i \in \mathbb{R}^d$ denote the i-th standard basis vector:

$$e_i = (0, \ldots, 0, \underbrace{1}_{i\text{-th position}}, 0, \ldots, 0).$$

All limits in W are taken with respect to $\|\cdot\|_W$.

Remark 26 (Why partial derivatives are stated on $\mathbb{R}^d$)**.** *Directional derivatives are intrinsic: for a map $f : \Omega \subseteq V \to W$ and a direction $v \in V$, the expression $t \mapsto a + tv$ makes sense in any normed vector space V, so $(D_v f)_a$ is defined without choosing coordinates. By contrast, partial derivatives are coordinate notions: the definition of $\partial_i f(a)$ uses the distinguished coordinate directions $e_1, \ldots, e_d$, which are canonically available only when the domain is identified with $\mathbb{R}^d$ (with its standard basis). If V is a general finite-dimensional normed space and one fixes an ordered basis $\beta = (v_1, \ldots, v_d)$ of V, then one may define a basis-dependent analogue by*

$$\partial_{v_i} f(a) := (D_{v_i} f)_a \qquad (i = 1, \ldots, d),$$

but this depends on the chosen basis and is not canonical.

Restricting f to a coordinate line

Proposition 19 (Local existence of coordinate line segments in Ω)**.** *Let $i \in \{1, \ldots, d\}$. Since Ω is open and $a \in \Omega$, there exists $r > 0$ such that*

$$B(a, r) \subseteq \Omega,$$

where $B(a, r)$ denotes the open ball in $\mathbb{R}^d$ (with its Euclidean norm). Set

$$I := (-r, r).$$

Then I is an interval containing 0 and

$$a + te_i \in \Omega \qquad \text{for all } t \in I.$$

Proof.
Let $r > 0$ satisfy $B(a, r) \subseteq \Omega$. If $t \in (-r, r)$, then

$$\|a + te_i - a\|_2 = \|te_i\|_2 = |t| < r,$$

so $a + te_i \in B(a, r) \subseteq \Omega$. □

Partial derivatives

Definition 16 (i-th partial derivative)**.** *Let $i \in \{1, \ldots, d\}$. Let I be an open interval containing 0 such that $a + te_i \in \Omega$ for all $t \in I$ (for instance, I as in Proposition 10). Define the one-variable map*

$$g_i : I \to W, \qquad g_i(t) := f(a + te_i).$$

If g_i is differentiable at 0, we define the i-th partial derivative of f at a by

$$\partial_i f(a) := g_i'(0).$$

Equivalently, if the limit exists in W, one has

$$\partial_i f(a) = \lim_{t \to 0} \frac{f(a + te_i) - f(a)}{t}.$$

Remark 27 (Interpretation)**.** *The partial derivative $\partial_i f(a)$ measures the first-order change of f at a when only the i-th coordinate is varied and all other coordinates are kept fixed. In particular, it is the derivative of the restriction of f to the coordinate line*

$$t \longmapsto a + te_i.$$

Connection to directional derivatives

Proposition 20 (Partial derivatives as special directional derivatives)**.** *Let $i \in \{1, \dots, d\}$. If the directional derivative $(D_{e_i} f)_a$ exists, then the i-th partial derivative exists and*

$$\partial_i f(a) = (D_{e_i} f)_a.$$

Conversely, if the i-th partial derivative $\partial_i f(a)$ exists, then the directional derivative in direction e_i exists and equals it.

Proof.
Both quantities are defined by the same one-variable limit:

$$\lim_{t \to 0} \frac{f(a + te_i) - f(a)}{t},$$

provided the limit exists in W. Hence they exist simultaneously and are equal. □

Connection to Fréchet differentiability

Proposition 21 (Fréchet differentiability implies existence of all partial derivatives)**.** *Assume f is Fréchet differentiable at a, with Fréchet derivative $(Df)_a : \mathbb{R}^d \to W$. Then for each $i \in \{1, \dots, d\}$ the partial derivative $\partial_i f(a)$ exists and satisfies*

$$\partial_i f(a) = (Df)_a(e_i).$$

Proof.
If f is Fréchet differentiable at a, then by the result proved in the section on directional derivatives, all directional derivatives exist and satisfy

$$(D_v f)_a = (Df)_a(v) \qquad (v \in \mathbb{R}^d).$$

Applying this with $v = e_i$ and using Proposition 11 yields

$$\partial_i f(a) = (D_{e_i} f)_a = (Df)_a(e_i).$$

□

Remark 28 (Partial derivatives are weaker hypiotheses than Fréchet differentiability)**.** *The existence of the partial derivatives $\partial_i f(a)$ for all i is a coordinate-axis condition. It does not, by itself, guarantee that f admits a single first-order linear approximation valid for all small increments $h \in \mathbb{R}^d$, and hence it does not, by itself, imply Fréchet differentiability at a.*

Differentiation Rules

Standing conventions

Throughout, $\mathbb{F}$ denotes $\mathbb{R}$. Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed $\mathbb{R}$-vector spaces, and let $\Omega \subseteq V$ be open. When $f : \Omega \to W$ is Fréchet differentiable at $a \in \Omega$, its derivative is denoted by $(Df)_a : V \to W$. All derivatives below are Fréchet derivatives.

Sum rule and scalar multiple rule

Proposition 22 (Sum rule)**.** *Let $f, g : \Omega \to W$ and let $a \in \Omega$. If f and g are Fréchet differentiable at a, then $f + g$ is Fréchet differentiable at a and*

$$(D(f+g))_a = (Df)_a + (Dg)_a,$$

where the sum on the right-hand side is the pointwise sum of linear maps $V \to W$.

Proof.
Let $f, g : \Omega \to W$ be Fréchet differentiable at a. By definition, there exist bounded linear maps $A : V \to W$ and $B : V \to W$ and maps

$$r_f : \{h \in V : a + h \in \Omega\} \to W, \qquad r_g : \{h \in V : a + h \in \Omega\} \to W$$

such that

$$f(a+h) = f(a) + A(h) + r_f(h), \qquad \lim_{h \to 0} \frac{\|r_f(h)\|_W}{\|h\|_V} = 0,$$

$$g(a+h) = g(a) + B(h) + r_g(h), \qquad \lim_{h \to 0} \frac{\|r_g(h)\|_W}{\|h\|_V} = 0.$$

Then for all h with $a + h \in \Omega$,

$$(f+g)(a+h) = (f+g)(a) + (A+B)(h) + \big(r_f(h) + r_g(h)\big).$$

Moreover,

$$\frac{\|r_f(h) + r_g(h)\|_W}{\|h\|_V} \leq \frac{\|r_f(h)\|_W}{\|h\|_V} + \frac{\|r_g(h)\|_W}{\|h\|_V} \longrightarrow 0.$$

Hence $f + g$ is Fréchet differentiable at a and $(D(f+g))_a = A + B = (Df)_a + (Dg)_a$. □

Proposition 23 (Scalar multiple rule)**.** *Let $f : \Omega \to W$, let $a \in \Omega$, and let $\lambda \in \mathbb{R}$. If f is Fréchet differentiable at a, then λf is Fréchet differentiable at a and*

$$(D(\lambda f))_a = \lambda (Df)_a.$$

Proof.
Let $f : \Omega \to W$ be Fréchet differentiable at a, and write

$$f(a+h) = f(a) + A(h) + r(h), \qquad \lim_{h \to 0} \frac{\|r(h)\|_W}{\|h\|_V} = 0,$$

where $A = (Df)_a$. Fix $\lambda \in \mathbb{R}$. Then

$$(\lambda f)(a+h) = (\lambda f)(a) + \lambda A(h) + \lambda r(h).$$

Since $\|\lambda r(h)\|_W = |\lambda| \, \|r(h)\|_W$, we have

$$\frac{\|\lambda r(h)\|_W}{\|h\|_V} = |\lambda| \frac{\|r(h)\|_W}{\|h\|_V} \longrightarrow 0.$$

Therefore λf is Fréchet differentiable at a and $(D(\lambda f))_a = \lambda A = \lambda (Df)_a$. □

Product rule

Proposition 24 (Product rule)**.** *Let $f, g : \Omega \to \mathbb{R}$ and let $a \in \Omega$. If f and g are Fréchet differentiable at a, then the pointwise product $fg : \Omega \to \mathbb{R}$ is Fréchet differentiable at a and*

$$(D(fg))_a = f(a)\,(Dg)_a + g(a)\,(Df)_a,$$

where the right-hand side is a linear map $V \to \mathbb{R}$ given by

$$\big(f(a)\,(Dg)_a + g(a)\,(Df)_a\big)(h) = f(a)\,(Dg)_a(h) + g(a)\,(Df)_a(h) \qquad (h \in V).$$

Proof.
Let $f, g : \Omega \to \mathbb{R}$ be Fréchet differentiable at a. Write

$$f(a+h) = f(a) + A(h) + r_f(h), \qquad \lim_{h\to 0} \frac{|r_f(h)|}{\|h\|_V} = 0,$$

$$g(a+h) = g(a) + B(h) + r_g(h), \qquad \lim_{h\to 0} \frac{|r_g(h)|}{\|h\|_V} = 0,$$

where $A = (Df)_a$ and $B = (Dg)_a$ (bounded linear maps $V \to \mathbb{R}$). For h with $a + h \in \Omega$,

$$\begin{aligned}
(fg)(a+h) &= f(a+h)\,g(a+h) \\
&= \big(f(a) + A(h) + r_f(h)\big)\big(g(a) + B(h) + r_g(h)\big) \\
&= f(a)g(a) + f(a)B(h) + g(a)A(h) \\
&\quad + \underbrace{A(h)B(h)}_{(*)} + \underbrace{f(a)r_g(h) + g(a)r_f(h)}_{(**)} + \underbrace{A(h)r_g(h) + B(h)r_f(h) + r_f(h)r_g(h)}_{(***)}.
\end{aligned}$$

Define the linear map $L : V \to \mathbb{R}$ by

$$L(h) := f(a)B(h) + g(a)A(h).$$

Then

$$(fg)(a+h) = (fg)(a) + L(h) + R(h),$$

where $R(h)$ is the sum of the terms $(*), (**), (***)$ above. We claim that

$$\frac{|R(h)|}{\|h\|_V} \to 0 \qquad (h \to 0).$$

Since A and B are bounded linear maps $V \to \mathbb{R}$, there exist constants $M_A, M_B \geq 0$ such that

$$|A(h)| \leq M_A\|h\|_V, \qquad |B(h)| \leq M_B\|h\|_V \qquad (h \in V).$$

For $(*)$ we have

$$\frac{|A(h)B(h)|}{\|h\|_V} \leq \frac{M_A\|h\|_V\, M_B\|h\|_V}{\|h\|_V} = M_A M_B\|h\|_V \to 0.$$

For $(**)$ we have

$$\frac{|f(a)r_g(h) + g(a)r_f(h)|}{\|h\|_V} \leq |f(a)|\frac{|r_g(h)|}{\|h\|_V} + |g(a)|\frac{|r_f(h)|}{\|h\|_V} \to 0.$$

For $(***)$ we have

$$\frac{|A(h)r_g(h)|}{\|h\|_V} \leq M_A|r_g(h)|, \qquad \frac{|B(h)r_f(h)|}{\|h\|_V} \leq M_B|r_f(h)|.$$

Since $r_f(h) = o(\|h\|_V)$ and $r_g(h) = o(\|h\|_V)$, in particular $r_f(h) \to 0$ and $r_g(h) \to 0$, hence both terms tend to 0. Finally,

$$\frac{|r_f(h) r_g(h)|}{\|h\|_V} = \left(\frac{|r_f(h)|}{\|h\|_V} \right) |r_g(h)| \longrightarrow 0 \cdot 0 = 0.$$

Thus $|R(h)|/\|h\|_V \to 0$, so fg is Fréchet differentiable at a and $(D(fg))_a = L = f(a)(Dg)_a + g(a)(Df)_a$. □

Proposition 25 (Topological Space Version-A bounded factor times a vanishing factor vanishes)**.** *Let X be a topological space, let $D \subseteq X$, and let a be an accumulation point of D. Let $u : D \to \mathbb{R}$ and $v : D \to \mathbb{R}$ be functions such that*

$$\lim_{x \to a} u(x) = 0,$$

and such that v is bounded in a neighbourhood of a, meaning: there exist a neighbourhood U of a in X and a constant $C > 0$ such that

$$x \in D \cap (U \setminus \{a\}) \quad \Longrightarrow \quad |v(x)| \leq C.$$

Then

$$\lim_{x \to a} u(x) v(x) = 0.$$

Proof.

Let $\varepsilon > 0$. Choose a neighbourhood U of a and $C > 0$ as in the boundedness assumption for v. Since $\lim_{x \to a} u(x) = 0$, there exists a neighbourhood U_1 of a such that

$$x \in D \cap (U_1 \setminus \{a\}) \quad \Longrightarrow \quad |u(x)| < \frac{\varepsilon}{C}.$$

Set $U_0 := U \cap U_1$, which is a neighbourhood of a. Then for all $x \in D \cap (U_0 \setminus \{a\})$ we have $|v(x)| \leq C$ and $|u(x)| < \varepsilon/C$, hence

$$|u(x)v(x)| = |u(x)|\,|v(x)| < \frac{\varepsilon}{C}\, C = \varepsilon.$$

This is the ε–neighbourhood condition for $\lim_{x \to a} u(x)v(x) = 0$. □

Proposition 26 (Metric Space Version-A bounded factor times a vanishing factor vanishes)**.** *Let (X, d) be a metric space, let $D \subseteq X$, and let a be an accumulation point of D. Let $u : D \to \mathbb{R}$ and $v : D \to \mathbb{R}$ be functions such that*

$$\lim_{x \to a} u(x) = 0,$$

and such that v is bounded near a, meaning: there exist $\delta > 0$ and $C > 0$ such that

$$x \in D, \quad 0 < d(x, a) < \delta \quad \Longrightarrow \quad |v(x)| \leq C.$$

Then

$$\lim_{x \to a} u(x) v(x) = 0.$$

Proof.
Let $\varepsilon > 0$. Choose $\delta > 0$ and $C > 0$ as in the boundedness assumption for v. Since $\lim_{x\to a} u(x) = 0$, there exists $\eta > 0$ such that for all $x \in D$,

$$0 < d(x,a) < \eta \quad \Longrightarrow \quad |u(x)| < \frac{\varepsilon}{C}.$$

Set $\rho := \min\{\delta, \eta\}$. Then for all $x \in D$ with $0 < d(x,a) < \rho$, we have $|v(x)| \le C$ and $|u(x)| < \varepsilon/C$, hence

$$|u(x)v(x)| = |u(x)|\,|v(x)| < \frac{\varepsilon}{C}\,C = \varepsilon.$$

This is the ε–ρ condition for $\lim_{x\to a} u(x)v(x) = 0$. □

Proposition 27 (Normed-Space Version-A bounded factor times a vanishing factor vanishes)**.** *Let $(V, \|\cdot\|_V)$ be a normed $\mathbb{R}$-vector space, let $D \subseteq V$, and let a be an accumulation point of D (with respect to the metric induced by $\|\cdot\|_V$). Let $u : D \to \mathbb{R}$ and $v : D \to \mathbb{R}$ be functions such that*

$$\lim_{x\to a} u(x) = 0,$$

and such that v is bounded near a, meaning: there exist $\delta > 0$ and $C > 0$ such that

$$x \in D, \quad 0 < \|x - a\|_V < \delta \quad \Longrightarrow \quad |v(x)| \le C.$$

Then

$$\lim_{x\to a} u(x)v(x) = 0.$$

Proof.
Let $\varepsilon > 0$. Choose $\delta > 0$ and $C > 0$ as in the boundedness assumption for v. Since $\lim_{x\to a} u(x) = 0$, there exists $\eta > 0$ such that for all $x \in D$,

$$0 < \|x - a\|_V < \eta \quad \Longrightarrow \quad |u(x)| < \frac{\varepsilon}{C}.$$

Set $\rho := \min\{\delta, \eta\}$. Then for all $x \in D$ with $0 < \|x - a\|_V < \rho$, we have $|v(x)| \le C$ and $|u(x)| < \varepsilon/C$, hence

$$|u(x)v(x)| = |u(x)|\,|v(x)| < \frac{\varepsilon}{C}\,C = \varepsilon.$$

This is the ε–ρ condition for $\lim_{x\to a} u(x)v(x) = 0$. □

Chain rule

Proposition 28 (Chain rule)**.** *Let $(Z, \|\cdot\|_Z)$ be a normed $\mathbb{R}$-vector space. Let $f : \Omega \to W$ and let $E \subseteq W$ be open with $f(\Omega) \subseteq E$. Let $g : E \to Z$. Fix $a \in \Omega$. If f is Fréchet differentiable at a and g is Fréchet differentiable at $f(a)$, then $g \circ f : \Omega \to Z$ is Fréchet differentiable at a and*

$$(D(g \circ f))_a = (Dg)_{f(a)} \circ (Df)_a,$$

where the right-hand side is the composition of linear maps $V \to W$ and $W \to Z$.

Proof.
Assume f is Fréchet differentiable at a and g is Fréchet differentiable at $f(a)$. Write

$$f(a+h) = f(a) + A(h) + r_f(h), \qquad \lim_{h\to 0} \frac{\|r_f(h)\|_W}{\|h\|_V} = 0,$$

where $A = (Df)_a$. Also, since g is Fréchet differentiable at $f(a)$, there exist a bounded linear map $B : W \to Z$ and a remainder $r_g : W \to Z$ (defined on a neighbourhood of 0 in W) such that for $k \to 0$ in W,

$$g(f(a)+k) = g(f(a)) + B(k) + r_g(k), \qquad \lim_{k\to 0} \frac{\|r_g(k)\|_Z}{\|k\|_W} = 0,$$

where $B = (Dg)_{f(a)}$.

For h sufficiently close to 0 we have $a + h \in \Omega$ and thus

$$(g \circ f)(a+h) = g(f(a+h)) = g\big(f(a) + A(h) + r_f(h)\big).$$

Apply the expansion for g with $k = A(h) + r_f(h)$:

$$(g \circ f)(a+h) = g(f(a)) + B(A(h) + r_f(h)) + r_g(A(h) + r_f(h)).$$

Using linearity of B gives

$$(g \circ f)(a+h) = (g \circ f)(a) + (B \circ A)(h) + \underbrace{B(r_f(h)) + r_g(A(h) + r_f(h))}_{=:R(h)}.$$

We show $\|R(h)\|_Z/\|h\|_V \to 0$. Since B is bounded, there exists $M \geq 0$ such that $\|B(w)\|_Z \leq M\|w\|_W$ for all $w \in W$, hence

$$\frac{\|B(r_f(h))\|_Z}{\|h\|_V} \leq M \frac{\|r_f(h)\|_W}{\|h\|_V} \to 0.$$

Next, since A is bounded there exists $M_A \geq 0$ with $\|A(h)\|_W \leq M_A\|h\|_V$, and since $r_f(h) = o(\|h\|_V)$ we have $\|r_f(h)\|_W \to 0$. Therefore $\|A(h)+r_f(h)\|_W \to 0$ as $h \to 0$. Using $\|r_g(k)\|_Z/\|k\|_W \to 0$ as $k \to 0$, we obtain

$$\frac{\|r_g(A(h) + r_f(h))\|_Z}{\|A(h) + r_f(h)\|_W} \to 0.$$

Moreover,

$$\frac{\|A(h) + r_f(h)\|_W}{\|h\|_V} \leq \frac{\|A(h)\|_W}{\|h\|_V} + \frac{\|r_f(h)\|_W}{\|h\|_V} \leq M_A + \frac{\|r_f(h)\|_W}{\|h\|_V},$$

so the ratio $\|A(h) + r_f(h)\|_W/\|h\|_V$ is bounded for h near 0. Hence, by proposition 18,

$$\frac{\|r_g(A(h) + r_f(h))\|_Z}{\|h\|_V} = \left(\frac{\|r_g(A(h) + r_f(h))\|_Z}{\|A(h) + r_f(h)\|_W}\right)\left(\frac{\|A(h) + r_f(h)\|_W}{\|h\|_V}\right) \to 0.$$

Therefore $\|R(h)\|_Z/\|h\|_V \to 0$, and $(g \circ f)$ is Fréchet differentiable at a with

$$(D(g \circ f))_a = B \circ A = (Dg)_{f(a)} \circ (Df)_a.$$

□

Remark 29 (Interpretation)**.** *The chain rule states that the first-order (linear) approximation of the composite map $g \circ f$ at a is obtained by composing the first-order approximations of f at a and g at $f(a)$.*

Quotient rule

Proposition 29 (Quotient rule)**.** *Let $f, g : \Omega \to \mathbb{R}$ and let $a \in \Omega$ with $g(a) \neq 0$. Assume f and g are Fréchet differentiable at a. Then there exists an open neighbourhood $U \subseteq \Omega$ of a such that $g(x) \neq 0$ for all $x \in U$, hence the quotient*

$$\frac{f}{g} : U \to \mathbb{R}, \qquad x \mapsto \frac{f(x)}{g(x)},$$

is well-defined. Moreover, f/g is Fréchet differentiable at a and

$$\left(D\left(\frac{f}{g}\right)\right)_a = \frac{g(a)\,(Df)_a - f(a)\,(Dg)_a}{(g(a))^2}.$$

Proof.
Let $f, g : \Omega \to \mathbb{R}$ be Fréchet differentiable at a with $g(a) \neq 0$. Since Fréchet differentiability implies continuity, g is continuous at a and therefore there exists an open neighbourhood $U \subseteq \Omega$ of a such that $g(x) \neq 0$ for all $x \in U$.

Define $h : (\mathbb{R} \setminus \{0\}) \to \mathbb{R}$ by $h(t) = 1/t$. The map h is differentiable at $g(a)$ with

$$(Dh)_{g(a)}(s) = -\frac{s}{(g(a))^2} \qquad (s \in \mathbb{R}),$$

i.e. $(Dh)_{g(a)} = -\frac{1}{(g(a))^2}\,\mathrm{id}_{\mathbb{R}}$. By the chain rule, the composition $h \circ g : U \to \mathbb{R}$ is Fréchet differentiable at a and

$$(D(h \circ g))_a = (Dh)_{g(a)} \circ (Dg)_a = -\frac{1}{(g(a))^2}(Dg)_a.$$

On U we have $h \circ g = 1/g$, hence $1/g$ is Fréchet differentiable at a and

$$\left(D\left(\frac{1}{g}\right)\right)_a = -\frac{1}{(g(a))^2}(Dg)_a.$$

Now on U we can write

$$\frac{f}{g} = f \cdot \frac{1}{g}.$$

Applying the product rule (already proved) to the real-valued functions f and $1/g$ yields

$$\left(D\left(\frac{f}{g}\right)\right)_a = f(a)\left(D\left(\frac{1}{g}\right)\right)_a + \frac{1}{g(a)}(Df)_a.$$

Substituting $(D(1/g))_a = -\frac{1}{(g(a))^2}(Dg)_a$ gives

$$\left(D\left(\frac{f}{g}\right)\right)_a = f(a)\left(-\frac{1}{(g(a))^2}(Dg)_a\right) + \frac{1}{g(a)}(Df)_a = \frac{g(a)\,(Df)_a - f(a)\,(Dg)_a}{(g(a))^2}.$$

□

Remark 30 (Why f/g is defined near a)**.** *Since g is continuous at a (Fréchet differentiability implies continuity) and $g(a) \neq 0$, there exists an open neighbourhood $U \subseteq \Omega$ of a such that $g(x) \neq 0$ for all $x \in U$.*

Graphs, Level Sets, Gradients, and Tangent Hyperplanes

Standing conventions

Throughout, $\mathbb{F}$ denotes $\mathbb{R}$. Let $(V, \|\cdot\|_V)$ be a normed $\mathbb{R}$-vector space, let $\Omega \subseteq V$ be open, and let $f : \Omega \to \mathbb{R}$ be a map. If f is Fréchet differentiable at $a \in \Omega$, then its derivative is denoted by $(Df)_a : V \to \mathbb{R}$. In the Euclidean case $V = \mathbb{R}^d$, we use the standard inner product $\langle x, y\rangle = \sum_{i=1}^d x_i y_i$ and the corresponding Euclidean norm $\|\cdot\|_2$.

Graph of a function

Definition 17 (Graph of a function)**.** *Let $f : \Omega \to \mathbb{R}$. The* graph *of f is the subset of $V \times \mathbb{R}$ defined by*

$$\operatorname{graph}(f) := \{(x, t) \in V \times \mathbb{R} : x \in \Omega,\ t = f(x)\}.$$

Level set

Definition 18 (Level set)**.** *Let $c \in \mathbb{R}$. The* level set *of f at level c is the subset of Ω defined by*

$$L_c := \{x \in \Omega : f(x) = c\} = f^{-1}(\{c\}).$$

If $a \in \Omega$, the level set through a *is $L_{f(a)}$.*

Tangent hyperplane to the graph

Definition 19 (Tangent hyperplane to the graph)**.** *Let $a \in \Omega$ and set $p := (a, f(a)) \in V \times \mathbb{R}$. Assume f is Fréchet differentiable at a. The* tangent hyperplane to $\operatorname{graph}(f)$ at p *is the affine subset of $V \times \mathbb{R}$ defined by*

$$T_p \operatorname{graph}(f) := \{(x, t) \in V \times \mathbb{R} : t = f(a) + (Df)_a(x - a)\}.$$

Remark 31 (Interpretation)**.** *The tangent hyperplane to the graph is the graph of the affine map*

$$x \longmapsto f(a) + (Df)_a(x - a),$$

which is the first-order (affine) approximation of f at a.

Tangent hyperplane to a level set

Definition 20 (Tangent hyperplane to a level set)**.** *Let $a \in \Omega$ and set $c := f(a)$, so that $a \in L_c$. Assume f is Fréchet differentiable at a. The* tangent hyperplane to the level set L_c at a *is the affine subset of V defined by*

$$T_a L_c := \{x \in V : (Df)_a(x - a) = 0\} = a + \ker\big((Df)_a\big).$$

Remark 32 (Interpretation)**.** *Fix $a \in \Omega$ and set $c := f(a)$, so the level set through a is*

$$L_c = \{x \in \Omega : f(x) = c\}.$$

A point $x \in L_c$ satisfies the constraint $f(x) = f(a)$, equivalently $f(x) - f(a) = 0$. Writing $x = a + h$, this becomes

$$f(a+h) - f(a) = 0.$$

Since f is Fréchet differentiable at a, there exists a remainder $r(h)$ such that

$$f(a+h) - f(a) = (Df)_a(h) + r(h), \qquad \frac{|r(h)|}{\|h\|_V} \to 0 \quad (h \to 0).$$

Thus, for small h, the quantity $f(a+h)-f(a)$ is well-approximated by the linear term $(Df)_a(h)$, and the remainder $r(h)$ is of smaller order than $\|h\|_V$. The equation $f(a+h) - f(a) = 0$ can therefore be viewed as

$$(Df)_a(h) + r(h) = 0,$$

and its linearisation at *a is obtained by discarding the higher-order term $r(h)$, yielding the linear condition*

$$(Df)_a(h) = 0.$$

The linearised constraint is $(Df)_a(h) = 0$, whose solution set in the increment variable h is

$$\ker((Df)_a) = \{h \in V : (Df)_a(h) = 0\}.$$

A point $x \in V$ lies on the tangent hyperplane to the level set L_c at a precisely when its increment from a, $h = x - a$, belongs to this kernel. Thus the tangent hyperplane is

$$\{x \in V : x - a \in \ker((Df)_a)\} = \{x \in V : (Df)_a(x-a) = 0\} = a + \ker((Df)_a).$$

It consists of exactly those directions h at a for which the first-order change of f vanishes, in the sense that $(Df)_a(h) = 0$ implies

$$f(a+h) = f(a) + o(\|h\|_V) \qquad (h \to 0 \text{ along the direction } h).$$

Gradient in $\mathbb{R}^d$

Definition 21 (Gradient)**.** *Assume $V = \mathbb{R}^d$ with the standard inner product $\cdot$. Let $f : \Omega \subseteq \mathbb{R}^d \to \mathbb{R}$ be Fréchet differentiable at $a \in \Omega$. The* gradient of f at *a is the unique vector $\nabla f(a) \in \mathbb{R}^d$ such that*

$$(Df)_a(h) = \langle \nabla f(a), h \rangle \qquad (h \in \mathbb{R}^d).$$

Remark 33 (Coordinate expression)**.** *If f has partial derivatives at a and is Fréchet differentiable at a, then*

$$\nabla f(a) = \big(\partial_1 f(a), \ldots, \partial_d f(a)\big).$$

Remark 34 (Level-set tangent hyperplane in terms of the gradient)**.** *Assume $V = \mathbb{R}^d$ and f is Fréchet differentiable at a. Then the tangent hyperplane to the level set $L_{f(a)}$ at a can be written as*

$$T_a L_{f(a)} = \{x \in \mathbb{R}^d : \langle \nabla f(a), (x-a) \rangle = 0\}.$$

In particular, $\nabla f(a)$ is normal to $T_a L_{f(a)}$.

The Operator Norm

Standing conventions

Throughout, $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ denote normed $\mathbb{R}$-vector spaces. We write $B(V, W)$ for the vector space of bounded linear maps $V \to W$. If $L \in B(V, W)$, then $\|L(v)\|_W$ denotes the norm of the vector $L(v) \in W$ (not a norm of L). All suprema below are taken in $\mathbb{R}$.

Upper bounds and supremum

Definition 22 (Upper bound and supremum)**.** *Let $S \subseteq \mathbb{R}$. A number $M \in \mathbb{R}$ is an* upper bound *for S if $s \leq M$ for all $s \in S$. If S is bounded above, its* supremum *is the real number* $\sup S$ *characterised by:*

1. $\sup S$ *is an upper bound for S, and*
2. *if U is any upper bound for S, then* $\sup S \leq U$.

Definition of the operator norm

Definition 23 (Operator norm)**.** *Let $L \in B(V, W)$. The* operator norm *of L is*

$$\|L\|_{V\to W} := \sup\{\|L(x)\|_W : \ x \in V, \ \|x\|_V \leq 1\}.$$

Remark 35 (Finite-dimensional case)**.** *If V is finite-dimensional, then the closed unit ball $\{x \in V : \|x\|_V \leq 1\}$ is compact and the map $x \mapsto \|L(x)\|_W$ is continuous, hence the supremum in the definition is attained. Therefore*

$$\|L\|_{V\to W} = \max\{\|L(x)\|_W : \ x \in V, \ \|x\|_V \leq 1\}.$$

Proposition 30 (Operator norm as a least constant)**.** *Let $L \in B(V, W)$. Then*

$$\|L\|_{V\to W} = \min\Big\{ C \in [0, \infty) : \|L(v)\|_W \leq C\,\|v\|_V \text{ for all } v \in V \Big\}.$$

Proposition 31 (Equivalent ratio formula)**.** *Let $L \in B(V, W)$. Then*

$$\|L\|_{V\to W} = \sup\left\{ \frac{\|L(v)\|_W}{\|v\|_V} : \ v \in V, \ v \neq 0 \right\}.$$

Proof.
Let $v \neq 0$ and set $u := v/\|v\|_V$, so that $\|u\|_V = 1$. By linearity,

$$L(v) = L(\|v\|_V u) = \|v\|_V\, L(u),$$

so

$$\frac{\|L(v)\|_W}{\|v\|_V} = \|L(u)\|_W.$$

Hence the set $\{\|L(u)\|_W : \ \|u\|_V = 1\}$ coincides with the set $\{\|L(v)\|_W/\|v\|_V : \ v \neq 0\}$. Taking suprema and using $\sup_{\|x\|_V \leq 1} \|L(x)\|_W = \sup_{\|x\|_V = 1} \|L(x)\|_W$ gives the claim. □

Remark 36. *The "worst" stretching occurs on the boundary of the unit ball*

Remark 37 (Why the definition uses the closed unit ball)**.** *Proposition 25 shows that the quantity*

$$\sup\left\{\frac{\|L(v)\|_W}{\|v\|_V} : v \in V,\ v \neq 0\right\}$$

can be computed by restricting to vectors of norm 1*, because every* $v \neq 0$ *can be written as* $v = \|v\|_V\, u$ *with* $\|u\|_V = 1$ *and then* $\|L(v)\|_W/\|v\|_V = \|L(u)\|_W$*. Since the unit sphere* $\{u \in V : \|u\|_V = 1\}$ *is contained in the closed unit ball* $\{x \in V : \|x\|_V \leq 1\}$*, one may equivalently define*

$$\|L\|_{V\to W} = \sup\{\|L(x)\|_W :\ \|x\|_V \leq 1\}.$$

Thus the use of the closed unit ball in the definition is a convenient normalisation: it collects all directions with input size at most 1 *into a single set, and it yields the same least upper bound as the ratio formula.*

Remark 38 (Conceptual meaning and relation between the two characterisations)**.** *Let* $L \in B(V, W)$*. By definition,*

$$\|L\|_{V\to W} = \sup\{\|L(x)\|_W :\ x \in V,\ \|x\|_V \leq 1\},$$

so $\|L\|_{V\to W}$ *is the least upper bound on the set of output norms produced by inputs of norm at most* 1*.*

For each nonzero $v \in V$*, the ratio*

$$\frac{\|L(v)\|_W}{\|v\|_V}$$

measures the norm expansion factor of L *in the direction* v*: it compares the size of the output vector* $L(v)$ *to the size of the input vector* v*. The operator norm* $\|L\|_{V\to W}$ *is the least upper bound of these expansion factors, that is,*

$$\frac{\|L(v)\|_W}{\|v\|_V} \leq \|L\|_{V\to W} \qquad (v \in V,\ v \neq 0).$$

Equivalently, $\|L\|_{V\to W}$ *is the smallest constant* $C \geq 0$ *such that*

$$\|L(v)\|_W \leq C\,\|v\|_V \qquad (v \in V),$$

which is the content of the least characterisation in the preceding proposition.

In this sense, $\|L\|_{V\to W}$ *is the maximal stretching factor of* L *with respect to the given norms: no vector is stretched by a factor larger than* $\|L\|_{V\to W}$*, and* $\|L\|_{V\to W}$ *is the least number with this property.*

Remark 39 (Why one may write min in finite dimensions)**.** *Let* $(V, \|\cdot\|_V)$ *be finite-dimensional and let* $L \in B(V, W)$*. Consider the set of constants*

$$\mathcal{C} := \Big\{\, C \in [0, \infty) : \|L(v)\|_W \leq C\,\|v\|_V \text{ for all } v \in V \,\Big\}.$$

In general normed spaces, one always has

$$\|L\|_{V\to W} = \inf \mathcal{C},$$

because $\|L\|_{V\to W}$ *is the least upper bound of the stretching factors and hence the least constant for which the estimate* $\|L(v)\|_W \leq C\|v\|_V$ *holds for all* v*.*

In finite dimensions, the unit sphere

$$S := \{v \in V : \|v\|_V = 1\}$$

is compact (Heine–Borel), and the function $\phi : S \to \mathbb{R}$ defined by

$$\phi(v) := \|L(v)\|_W$$

is continuous. By the extreme value theorem, ϕ attains a maximum at some $v_0 \in S$. Setting

$$C_0 := \phi(v_0) = \|L(v_0)\|_W,$$

one has $\|L(v)\|_W \leq C_0\|v\|_V$ for all $v \in V$, and no smaller constant works because equality holds for $v = v_0$. Hence the infimum of $\mathcal{C}$ is attained, and therefore

$$\|L\|_{V\to W} = \min \mathcal{C}.$$

Basic inequality for the operator norm

Proposition 32 (Basic inequality for the operator norm)**.** *Let $L \in B(V, W)$. Then for every $v \in V$,*

$$\|L(v)\|_W \leq \|L\|_{V\to W}\, \|v\|_V.$$

Proof.
If $v = 0$, then $\|L(v)\|_W = 0$ and the inequality holds. Assume $v \neq 0$ and set $u := v/\|v\|_V$, so that $\|u\|_V = 1$. Then $v = \|v\|_V u$ and by linearity

$$L(v) = L(\|v\|_V u) = \|v\|_V\, L(u),$$

hence

$$\|L(v)\|_W = \|v\|_V\, \|L(u)\|_W.$$

Since $\|u\|_V \leq 1$, the value $\|L(u)\|_W$ belongs to the set $\{\|L(x)\|_W : \ \|x\|_V \leq 1\}$, hence by the defining property of supremum,

$$\|L(u)\|_W \leq \sup\{\|L(x)\|_W : \ \|x\|_V \leq 1\} = \|L\|_{V\to W}.$$

Multiplying by $\|v\|_V$ gives $\|L(v)\|_W \leq \|L\|_{V\to W}\, \|v\|_V$. □

Remark 40 (Interpretation)**.** *Proposition 26 states that the operator norm $\|L\|_{V\to W}$ is an upper bound on all norm expansion factors of L: for every nonzero $v \in V$,*

$$\frac{\|L(v)\|_W}{\|v\|_V} \leq \|L\|_{V\to W}.$$

Equivalently, Proposition 26 states that the same constant $\|L\|_{V\to W}$ bounds $\|L(v)\|_W$ for every input vector $v \in V$, namely

$$\|L(v)\|_W \leq \|L\|_{V\to W}\, \|v\|_V \qquad (v \in V).$$

(the output norm cannot exceed (maximum stretch factor)×(input norm))

Jacobian and Hessian

Standing conventions

Let $\Omega \subseteq \mathbb{R}^d$ be open. All norms on $\mathbb{R}^d$ and $\mathbb{R}^m$ are taken to be the Euclidean norms, and all derivatives are Fréchet derivatives.

The Jacobian matrix

Let $f : \Omega \to \mathbb{R}^m$ be Fréchet differentiable at $a \in \Omega$. Then the derivative $(Df)_a : \mathbb{R}^d \to \mathbb{R}^m$ is a bounded linear map. The *Jacobian matrix* of f at a is the unique matrix $J_f(a) \in \mathbb{R}^{m\times d}$ such that

$$(Df)_a(h) = J_f(a)\, h \qquad (h \in \mathbb{R}^d),$$

where the right-hand side is ordinary matrix–vector multiplication.

Remark 41 (Conceptual meaning)**.** *The Jacobian is the coordinate representation of the first-order linear approximation. For small increments h, one has*

$$f(a+h) - f(a) \approx (Df)_a(h) = J_f(a)\, h.$$

Equivalently, the j-th column of $J_f(a)$ is $(Df)_a(e_j)$, i.e. the first-order change in the output produced by moving in the j-th coordinate direction.

Remark 42 (Entries via component functions)**.** *Write $f = (f_1, \dots, f_m)$ with $f_i : \Omega \to \mathbb{R}$. If f is Fréchet differentiable at a, then each partial derivative $\partial_j f_i(a)$ exists and*

$$J_f(a) = \big(\partial_j f_i(a)\big)_{1\le i\le m,\ 1\le j\le d}.$$

The Hessian matrix (scalar-valued maps)

Let $f : \Omega \to \mathbb{R}$ be twice Fréchet differentiable at $a \in \Omega$. Then the second derivative $(D^2 f)_a : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ is a bounded bilinear map. The *Hessian matrix* of f at a is the unique matrix $H_f(a) \in \mathbb{R}^{d\times d}$ such that

$$(D^2 f)_a(h,k) = h^\top H_f(a)\, k \qquad (h,k \in \mathbb{R}^d).$$

In particular,

$$(D^2 f)_a(h,h) = h^\top H_f(a)\, h \qquad (h \in \mathbb{R}^d).$$

Remark 43 (Conceptual meaning)**.** *The Hessian is the coordinate representation of the second-order behaviour of f at a. It governs the quadratic term in the Taylor polynomial: if f admits a Taylor expansion of order 2 at a, then*

$$f(a+h) = f(a) + (Df)_a(h) + \frac{1}{2}(D^2 f)_a(h,h) + \textit{(terms negligible relative to } \|h\|_2^2\textit{)}.$$

Thus $H_f(a)$ describes the quadratic correction to the linear approximation.

Remark 44 (Entries via second partial derivatives)**.** *If the second-order partial derivatives exist at a (and the standard identification applies), then*

$$H_f(a) = \big(\partial_i \partial_j f(a)\big)_{1\le i,j\le d}.$$

The Mean Value Inequality

Standing conventions

Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed $\mathbb{R}$-vector spaces. All limits in W are taken with respect to $\|\cdot\|_W$, and all suprema below are taken in $\mathbb{R}$.

If $I \subseteq \mathbb{R}$ is an open interval and $g : I \to W$ is differentiable, then $g'(t) \in W$ denotes its one-variable derivative at $t \in I$.

If $\Omega \subseteq V$ is open and $f : \Omega \to W$ is Fréchet differentiable at $x \in \Omega$, then its derivative is denoted by

$$(Df)_x \in B(V, W),$$

and its operator norm by $\|(Df)_x\|_{V \to W}$.

The mean value inequality in one variable

Proposition 33 (Mean value inequality)**.** *Let $[a, b] \subseteq \mathbb{R}$ with $a < b$, and let $g : [a, b] \to W$. Assume that g is continuous on $[a, b]$ and differentiable on (a, b). Assume moreover that the set*

$$S := \{\|g'(t)\|_W : t \in (a, b)\}$$

is bounded above in $\mathbb{R}$. Then

$$\|g(b) - g(a)\|_W \leq (b - a) \sup_{t \in (a,b)} \|g'(t)\|_W.$$

Equivalently, if $M \geq 0$ is any number satisfying

$$\|g'(t)\|_W \leq M \qquad \text{for all } t \in (a, b),$$

then

$$\|g(b) - g(a)\|_W \leq M\,(b - a).$$

Remark 45 (Interpretation)**.** *The quantity $\|g'(t)\|_W$ is the speed of the curve $t \mapsto g(t)$ in the normed space W. If the speed is bounded above on (a, b), then the total displacement from $g(a)$ to $g(b)$ cannot exceed*

$$\text{(maximal speed)} \times \text{(time interval length)}.$$

Remark 46 (Why a supremum appears)**.** *The number $\sup_{t \in (a,b)} \|g'(t)\|_W$ is the least upper bound of the set $S = \{\|g'(t)\|_W : t \in (a, b)\}$. Any upper bound M for S also yields a valid estimate, and one has*

$$\sup_{t \in (a,b)} \|g'(t)\|_W \leq M.$$

Thus using the supremum gives the sharpest constant available from this information.

Line segments and a reduction to one variable

Definition 24 (Line segment between two points)**.** *Let $a, b \in V$. The (closed) line segment from a to b is*

$$[a, b] := \{(1 - t)a + tb : \ t \in [0, 1]\}.$$

Definition 25 (Parametrisation of the segment)**.** *Let $a, b \in V$ and define*

$$\gamma : [0, 1] \to V, \qquad \gamma(t) := (1 - t)a + tb.$$

Then $\gamma([0, 1]) = [a, b]$ and γ is differentiable on $(0, 1)$ with

$$\gamma'(t) = b - a \qquad (t \in (0, 1)).$$

Remark 47 (What it means to parametrise the segment)**.** *To parametrise a subset $S \subseteq V$ means to describe its points as the image of a map from a simple parameter set. Concretely, it means to exhibit a map $\gamma : D \to V$ from a parameter set $D \subseteq \mathbb{R}$ such that*

$$\gamma(D) = S.$$

In particular, γ describes every point of S as $\gamma(t)$ for some parameter value t, and conversely every $\gamma(t)$ lies in S. For the line segment,

$$[a, b] = \{(1 - t)a + tb : \ t \in [0, 1]\},$$

the map $\gamma : [0, 1] \to V$ given by $\gamma(t) = (1 - t)a + tb$ satisfies $\gamma([0, 1]) = [a, b]$, with $\gamma(0) = a$ and $\gamma(1) = b$.

Remark 48 (Reduction to a one-variable map)**.** *Assume $[a, b] \subseteq \Omega$ and define*

$$g : [0, 1] \to W, \qquad g(t) := f(\gamma(t)).$$

Then

$$g(0) = f(a), \qquad g(1) = f(b),$$

so estimating $\|f(b) - f(a)\|_W$ is equivalent to estimating $\|g(1) - g(0)\|_W$.

Remark 49 (Idea of the reduction)**.** *The reduction is that the change from a to b in the domain V is described by a one-variable path $\gamma : [0, 1] \to V$ with $\gamma(0) = a$ and $\gamma(1) = b$. Composing with f gives a one-variable map $g = f \circ \gamma : [0, 1] \to W$ with $g(0) = f(a)$ and $g(1) = f(b)$, so*

$$\|f(b) - f(a)\|_W = \|g(1) - g(0)\|_W.$$

Mean value inequality along a segment

Proposition 34 (Mean value inequality along a line segment)**.** *Let $a, b \in \Omega$ and assume $[a, b] \subseteq \Omega$. Assume that f is Fréchet differentiable at every point of $[a, b]$. Assume that the set*

$$\left\{ \ \|(Df)_x\|_{V \to W} : \ x \in [a, b] \ \right\}$$

is bounded above in $\mathbb{R}$. Then

$$\|f(b) - f(a)\|_W \leq \left(\sup_{x \in [a,b]} \|(Df)_x\|_{V \to W} \right) \|b - a\|_V.$$

Equivalently, if $M \geq 0$ satisfies $\|(Df)_x\|_{V \to W} \leq M$ for all $x \in [a, b]$, then

$$\|f(b) - f(a)\|_W \leq M \, \|b - a\|_V.$$

Proof.

Define $\gamma(t) = (1 - t)a + tb$ and $g(t) = f(\gamma(t))$ as above. Then $g(0) = f(a)$ and $g(1) = f(b)$. Since f is Fréchet differentiable on $[a, b]$, the chain rule gives that g is differentiable on $(0, 1)$ and

$$g'(t) = (Df)_{\gamma(t)}(\gamma'(t)) \qquad (t \in (0, 1)).$$

Because $\gamma'(t) = b - a$, we have

$$g'(t) = (Df)_{\gamma(t)}(b - a) \qquad (t \in (0, 1)).$$

Taking norms and using the basic operator norm inequality $\|L(v)\|_W \le \|L\|_{V\to W}\|v\|_V$ yields

$$\|g'(t)\|_W = \|(Df)_{\gamma(t)}(b-a)\|_W \le \|(Df)_{\gamma(t)}\|_{V\to W}\,\|b-a\|_V \qquad (t \in (0,1)).$$

Set

$$M := \sup_{x\in[a,b]} \|(Df)_x\|_{V\to W}.$$

Then $\|(Df)_{\gamma(t)}\|_{V\to W} \le M$ for all $t \in (0,1)$, hence

$$\|g'(t)\|_W \le M\,\|b-a\|_V \qquad (t \in (0,1)).$$

Applying the one-variable mean value inequality to $g : [0,1] \to W$ gives

$$\|g(1) - g(0)\|_W \le M\,\|b-a\|_V.$$

Since $g(1) - g(0) = f(b) - f(a)$, this is the desired inequality. □

Multilinear Maps and Higher Fréchet Derivatives

Standing conventions

Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed $\mathbb{R}$-vector spaces and let $\Omega \subseteq V$ be open. We write $B(V,W)$ for the vector space of bounded linear maps $V \to W$ (with the operator norm).

Definition 26 (k-linear (multilinear) map)**.** *Let $k \in \mathbb{N}$ and let $V_1, \dots, V_k$ and W be $\mathbb{R}$-vector spaces. A map*

$$T : V_1 \times \cdots \times V_k \to W$$

is called k-linear *(or* multilinear*) if, for each $i \in \{1, \dots, k\}$, the map*

$$V_i \to W, \qquad x \mapsto T(v_1, \dots, v_{i-1}, x, v_{i+1}, \dots, v_k),$$

is linear for every fixed choice of $v_j \in V_j$ with $j \neq i$.

First derivative: pointwise derivative and derivative map

Remark 50 (Pointwise derivative)**.** *Let $f : \Omega \to W$ and let $a \in \Omega$. If f is Fréchet differentiable at a, then its pointwise derivative is a bounded linear map*

$$(Df)_a : V \to W.$$

The input of $(Df)_a$ is an increment *$h \in V$ (a direction/step from the base point a), and $(Df)_a(h)$ is the linear approximation to the change $f(a+h) - f(a)$ for small h. The condition $a + h \in \Omega$ is only needed to ensure that $f(a+h)$ is defined; it does not change the fact that $(Df)_a$ is a linear map on V.*

Remark 51 (Derivative map)**.** *Assume f is Fréchet differentiable at every point of Ω. The* derivative map *is the function*

$$Df : \Omega \to B(V,W), \qquad a \mapsto (Df)_a.$$

Thus Df takes a base point $a \in \Omega$ and returns the linearisation $(Df)_a$ of f at a.

Second derivative: three related objects

Definition 27 (Twice Fréchet differentiable at a point)**.** *Let $f : \Omega \to W$ be Fréchet differentiable on Ω and let $a \in \Omega$. We say that f is* twice Fréchet differentiable at *a if the derivative map*

$$Df : \Omega \to B(V, W)$$

is Fréchet differentiable at a (as a map into the normed space $B(V, W)$).

Remark 52 (Second derivative map)**.** *If f is twice Fréchet differentiable at every point of Ω, then Df is Fréchet differentiable on Ω and its derivative map is*

$$D(Df) : \Omega \to B\big(V, B(V, W)\big), \qquad a \mapsto (D(Df))_a.$$

Thus $D(Df)$ takes a base point $a \in \Omega$ and returns the pointwise derivative $(D(Df))_a$.

Remark 53 (Second pointwise derivative)**.** *If f is twice Fréchet differentiable at a, then the pointwise derivative of Df at a is a bounded linear map*

$$(D(Df))_a : V \to B(V, W).$$

Its input is an increment $h \in V$ (moving the base point from a to $a + h$), and its output $(D(Df))_a(h)$ is a bounded linear map $V \to W$.

Definition 28 (Second derivative as a bilinear map)**.** *Assume f is twice Fréchet differentiable at a. Define*

$$(D^2 f)_a : V \times V \to W$$

by

$$(D^2 f)_a(h, k) := \big((D(Df))_a(h)\big)(k) \qquad (h, k \in V).$$

Proposition 35 (Relation between $(D(Df))_a$ and $(D^2 f)_a$)**.** *Assume f is twice Fréchet differentiable at a. Then $(D^2 f)_a : V \times V \to W$ is bilinear, and for all $h, k \in V$,*

$$(D^2 f)_a(h, k) = \big((D(Df))_a(h)\big)(k).$$

Conversely, for each $h \in V$,

$$(D(Df))_a(h) : V \to W, \qquad k \mapsto (D^2 f)_a(h, k).$$

Proof.

Fix $h \in V$. The map $(D(Df))_a(h) \in B(V, W)$ is linear in k, hence $k \mapsto (D^2 f)_a(h, k)$ is linear. Fix $k \in V$. Since $(D(Df))_a : V \to B(V, W)$ is linear and evaluation at k is linear, the map $h \mapsto (D^2 f)_a(h, k)$ is linear. Thus $(D^2 f)_a$ is bilinear. The displayed identities follow directly from the definition of $(D^2 f)_a$. □

Higher derivatives

Definition 29 (Bounded n-linear maps)**.** *Let $n \in \mathbb{N}$. A map $T : V^n \to W$ is called* bounded n-linear *if it is n-linear and there exists $C \geq 0$ such that*

$$\|T(v_1, \ldots, v_n)\|_W \leq C \, \|v_1\|_V \cdots \|v_n\|_V \qquad (v_1, \ldots, v_n \in V).$$

The vector space of bounded n-linear maps $V^n \to W$ is denoted by $B_n(V, W)$.

Definition 30 (n-times Fréchet differentiable at a point)**.** *Let $n \in \mathbb{N}$ and let $f : \Omega \to W$. We define inductively what it means for f to be n-times Fréchet differentiable at $a \in \Omega$. For $n = 1$ this is the usual Fréchet differentiability. For $n \geq 2$, we say that f is* n-times Fréchet differentiable at *a if f is $(n-1)$-times Fréchet differentiable on a neighbourhood of a and the $(n-1)$-st derivative map*

$$D^{n-1}f : \Omega \to B_{n-1}(V, W), \qquad x \mapsto (D^{n-1}f)_x,$$

is Fréchet differentiable at a (as a map into the normed space $B_{n-1}(V, W)$).

Remark 54 (The three objects for the n-th derivative)**.** *Assume f is n-times Fréchet differentiable on Ω, with $n \geq 2$. Then the following objects are defined:*

1. *The* n-th derivative map *(an assignment of n-th derivatives to base points)*

$$D^n f : \Omega \to B_n(V, W), \qquad a \mapsto (D^n f)_a.$$

2. *The* n-th pointwise derivative *of the map $D^{n-1}f$ at a,*

$$\big(D(D^{n-1}f)\big)_a : V \to B_{n-1}(V, W).$$

3. *The* n-th derivative at *a as an n-linear map,*

$$(D^n f)_a : V^n \to W,$$

defined from the pointwise derivative by

$$(D^n f)_a(h_1, \ldots, h_n) = \Big(\big(D(D^{n-1}f)\big)_a(h_1)\Big)(h_2, \ldots, h_n) \qquad (h_1, \ldots, h_n \in V).$$

Proposition 36 (Multilinearity of $(D^n f)_a$)**.** *Let $n \geq 2$ and assume f is n-times Fréchet differentiable at a. Then $(D^n f)_a : V^n \to W$ defined above is n-linear. Moreover, for each $h_1 \in V$,*

$$\big(D(D^{n-1}f)\big)_a(h_1) : V^{n-1} \to W, \qquad (h_2, \ldots, h_n) \mapsto (D^n f)_a(h_1, \ldots, h_n).$$

Proof.
Fix $h_1 \in V$. Then $\big(D(D^{n-1}f)\big)_a(h_1) \in B_{n-1}(V, W)$ is multilinear in $(h_2, \ldots, h_n)$, hence the map $(h_2, \ldots, h_n) \mapsto (D^n f)_a(h_1, \ldots, h_n)$ is multilinear. Next, for fixed $(h_2, \ldots, h_n)$, the map $h_1 \mapsto \big(D(D^{n-1}f)\big)_a(h_1)$ is linear in $B_{n-1}(V, W)$, and evaluation at $(h_2, \ldots, h_n)$ is linear, so $h_1 \mapsto (D^n f)_a(h_1, \ldots, h_n)$ is linear. Therefore $(D^n f)_a$ is n-linear, and the displayed identity follows from the definition. □

Higher-Order Directional Derivatives

Standing conventions

Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed $\mathbb{R}$-vector spaces, let $\Omega \subseteq V$ be open, and let $f : \Omega \to W$ be a map. Fix $a \in \Omega$. Directional derivatives are taken in the sense defined earlier:

$$(D_v f)_a = \lim_{t \to 0} \frac{f(a + tv) - f(a)}{t},$$

when the limit exists in W.

Iterated directional derivatives

Definition 31 (n-th directional derivative)**.** *Let $n \in \mathbb{N}$ and let $v_1, \dots, v_n \in V$. Define iteratively, whenever the expressions exist:*

1. *$D_{v_1} f : \Omega \to W$ by $x \mapsto (D_{v_1} f)_x$;*
2. *for $k \geq 2$, define $D_{v_k} \cdots D_{v_1} f : \Omega \to W$ by*

$$x \mapsto \big(D_{v_k}(D_{v_{k-1}} \cdots D_{v_1} f)\big)_x.$$

If the resulting map $D_{v_n} \cdots D_{v_1} f$ is defined on a neighbourhood of a and is directionally differentiable at a as required by the iteration above, we define the n-th directional derivative of f at a in directions $v_1, \dots, v_n$ *by*

$$\big(D_{v_n} \cdots D_{v_1} f\big)_a \in W.$$

Remark 55 (Conceptual meaning)**.** *At each step one forms a new function by taking a first-order approximation along a line. Then one takes a first-order approximation of that new function along another line. Iterating this procedure produces higher-order directional derivatives.*

Connection to higher Fréchet derivatives

Proposition 37 (n-times Fréchet differentiable implies existence of n-th directional derivatives)**.** *Let $n \in \mathbb{N}$ and assume that f is n-times Fréchet differentiable at a. Then for every $v_1, \dots, v_n \in V$, the iterated directional derivative $\big(D_{v_n} \cdots D_{v_1} f\big)_a$ exists. Moreover,*

$$\big(D_{v_n} \cdots D_{v_1} f\big)_a = (D^n f)_a(v_n, \dots, v_1),$$

with the order of the arguments as written.

Remark 56 (Why the converse can fail)**.** *Even if all iterated directional derivatives $\big(D_{v_n} \cdots D_{v_1} f\big)_a$ exist (for many or all choices of directions), this does not by itself imply that f is n-times Fréchet differentiable at a. Fréchet differentiability requires a single bounded n-linear map $(D^n f)_a : V^n \to W$ together with a uniform n-th order approximation valid for all small increments in V. Directional derivatives are obtained by restricting to lines and then iterating, so they provide weaker information and need not assemble into such a single multilinear approximation.*

Remark 57 (Meaning of "uniform" in Fréchet differentiability)**.** *In the definition of Fréchet differentiability at a, the approximation is* uniform in the increment h*: for every $\varepsilon > 0$ there exists $\delta > 0$ such that the estimate*

$$\|f(a+h) - f(a) - (Df)_a(h)\|_W \leq \varepsilon \, \|h\|_V$$

holds for all $h \in V$ *with $0 < \|h\|_V < \delta$ and $a + h \in \Omega$. Thus, once ε is fixed, the same δ works simultaneously for every direction of h.*

Repeated directions

Remark 58 (Repeated direction)**.** *Let $v \in V$ and let $n \in \mathbb{N}$. If one repeats the same direction, then the iterated directional derivative is denoted by*

$$(D_v^n f)_a := (D_v \cdots D_v f)_a.$$

Let $I \subseteq \mathbb{R}$ be an open interval containing 0 such that $a + tv \in \Omega$ for all $t \in I$ (for example, choose I as in Proposition 8). Define

$$\gamma : I \to V, \qquad \gamma(t) = a + tv,$$

and define the one-variable map

$$g : I \to W, \qquad g(t) = f(\gamma(t)) = f(a + tv).$$

If the one-variable derivative $g^{(n)}(0)$ exists in W, then

$$(D_v^n f)_a = g^{(n)}(0).$$

Higher-Order Partial Derivatives

Standing conventions

Throughout, $d \in \mathbb{N}$ and $\Omega \subseteq \mathbb{R}^d$ is open (with the Euclidean norm). Let $(W, \|\cdot\|_W)$ be a normed $\mathbb{R}$-vector space and let $f : \Omega \to W$ be a map. Fix $a = (a_1, \ldots, a_d) \in \Omega$. For $i \in \{1, \ldots, d\}$, let $e_i \in \mathbb{R}^d$ denote the i-th standard basis vector. All limits in W are taken with respect to $\|\cdot\|_W$.

Iterated partial derivatives

Definition 32 (First-order partial derivatives as a family of functions)**.** *Let $i \in \{1, \ldots, d\}$. Assume that $\partial_i f(x)$ exists for every x in some neighbourhood of a (in particular, for $x = a$). Then we obtain a map*

$$\partial_i f : \Omega \to W, \qquad x \mapsto \partial_i f(x),$$

defined at least on that neighbourhood.

Definition 33 (Higher-order (iterated) partial derivative at a point)**.** *Let $n \in \mathbb{N}$ and let $i_1, \ldots, i_n \in \{1, \ldots, d\}$. Assume inductively that the map*

$$\partial_{i_{n-1}} \cdots \partial_{i_1} f$$

is defined on a neighbourhood of a and that its i_n-th partial derivative at a exists. Then we define the n-th order iterated partial derivative of f at a *by*

$$\partial_{i_n} \cdots \partial_{i_1} f(a) := \partial_{i_n}\big(\partial_{i_{n-1}} \cdots \partial_{i_1} f\big)(a) \in W.$$

Remark 59 (Order convention)**.** *The expression $\partial_{i_n} \cdots \partial_{i_1} f(a)$ is an* iterated *derivative: one first forms the function $\partial_{i_1} f$, then differentiates that function in the i_2-direction, and so on, finally taking the i_n-th partial derivative at a. No commutation of partial derivatives is asserted by this notation.*

Partial derivatives as directional derivatives

Proposition 38 (Higher-order partial derivatives as iterated directional derivatives)**.** *Let $n \in \mathbb{N}$ and let $i_1, \dots, i_n \in \{1, \dots, d\}$. Assume the iterated directional derivative*

$$\left(D_{e_{i_n}} \cdots D_{e_{i_1}} f\right)_a$$

exists (in the sense of the section on higher-order directional derivatives). Then the iterated partial derivative $\partial_{i_n} \cdots \partial_{i_1} f(a)$ exists and

$$\partial_{i_n} \cdots \partial_{i_1} f(a) = \left(D_{e_{i_n}} \cdots D_{e_{i_1}} f\right)_a.$$

Conversely, if $\partial_{i_n} \cdots \partial_{i_1} f(a)$ exists (with the neighbourhood hypotheses needed for the iteration), then the iterated directional derivative above exists and the same identity holds.

Remark 60 (Interpretation)**.** *Higher-order partial derivatives are obtained by restricting to coordinate lines one step at a time. Equivalently, they are higher-order directional derivatives along the distinguished directions $e_1, \dots, e_d$.*

Connection to higher Fréchet derivatives

Proposition 39 (n-times Fréchet differentiable implies existence of all n-th order partial derivatives)**.** *Let $n \in \mathbb{N}$ and assume that f is n-times Fréchet differentiable at a. Then for every choice of indices $i_1, \dots, i_n \in \{1, \dots, d\}$, the iterated partial derivative $\partial_{i_n} \cdots \partial_{i_1} f(a)$ exists and satisfies*

$$\partial_{i_n} \cdots \partial_{i_1} f(a) = (D^n f)_a(e_{i_n}, \dots, e_{i_1}),$$

with the order of the arguments as written.

Remark 61 (Coordinate expansion via multilinearity)**.** *Assume f is n-times Fréchet differentiable at $a \in \Omega$, so that $(D^n f)_a : (\mathbb{R}^d)^n \to W$ is an n-linear map. Let $v_1, \dots, v_n \in \mathbb{R}^d$. For each $r \in \{1, \dots, n\}$ write v_r in the standard basis as*

$$v_r = \sum_{j=1}^{d} (v_r)_j \, e_j,$$

where $(v_r)_j \in \mathbb{R}$ denotes the j-th coordinate of v_r. We expand $(D^n f)_a(v_1, \dots, v_n)$ by using linearity in each slot, one slot at a time.

First, fixing $v_2, \dots, v_n$ and using linearity in the first variable,

$$(D^n f)_a(v_1, v_2, \dots, v_n) = \sum_{j_1=1}^{d} (v_1)_{j_1} \, (D^n f)_a(e_{j_1}, v_2, \dots, v_n).$$

Next, for each term in this sum, fix $e_{j_1}, v_3, \dots, v_n$ and use linearity in the second variable:

$$(D^n f)_a(e_{j_1}, v_2, v_3, \dots, v_n) = \sum_{j_2=1}^{d} (v_2)_{j_2} \, (D^n f)_a(e_{j_1}, e_{j_2}, v_3, \dots, v_n).$$

Continuing in the same way through the n-th variable, we obtain the full expansion

$$(D^n f)_a(v_1, \dots, v_n) = \sum_{j_1, \dots, j_n=1}^{d} (v_1)_{j_1} \cdots (v_n)_{j_n} \, (D^n f)_a(e_{j_1}, \dots, e_{j_n}).$$

Finally, by the proposition relating higher Fréchet derivatives and iterated partial derivatives (with the fixed order convention),

$$(D^n f)_a(e_{j_1}, \dots, e_{j_n}) = \partial_{j_1} \cdots \partial_{j_n} f(a),$$

so the expansion can be written equivalently as

$$(D^n f)_a(v_1, \dots, v_n) = \sum_{j_1,\dots,j_n=1}^{d} (v_1)_{j_1} \cdots (v_n)_{j_n} \, \partial_{j_1} \cdots \partial_{j_n} f(a).$$

Symmetry of Mixed Partial Derivatives

Remark 62 (No automatic symmetry)**.** *Even in finite dimensions, the existence of the iterated partial derivatives does not by itself imply that*

$$\partial_i \partial_j f(a) = \partial_j \partial_i f(a)$$

or any higher-order commutation identity. Such symmetry requires additional hypotheses.

Proposition 40 (When iterated partial derivatives are symmetric)**.** *Let $\Omega \subseteq \mathbb{R}^d$ be open and let $f : \Omega \to W$, where $(W, \|\cdot\|_W)$ is a normed $\mathbb{R}$-vector space. Fix $a \in \Omega$ and an integer $n \geq 2$.*

Assume that f is n-times Fréchet differentiable on a neighbourhood of a and that the n-th derivative map

$$D^n f : \Omega \to B_n(\mathbb{R}^d, W), \qquad x \mapsto (D^n f)_x,$$

is continuous at a (equivalently: f is of class C^n in a neighbourhood of a). Then the n-linear map $(D^n f)_a : (\mathbb{R}^d)^n \to W$ is symmetric*, meaning that for every permutation σ of $\{1, \dots, n\}$ and every $v_1, \dots, v_n \in \mathbb{R}^d$ one has*

$$(D^n f)_a(v_1, \dots, v_n) = (D^n f)_a\big(v_{\sigma(1)}, \dots, v_{\sigma(n)}\big).$$

In particular, for any indices $i_1, \dots, i_n \in \{1, \dots, d\}$ and any permutation σ,

$$\partial_{i_n} \cdots \partial_{i_1} f(a) = \partial_{i_{\sigma(n)}} \cdots \partial_{i_{\sigma(1)}} f(a).$$

Warning. *Without a continuity hypothesis on the relevant higher derivatives in a neighbourhood of a, iterated partial derivatives need not commute: it can happen that $\partial_i \partial_j f(a) \neq \partial_j \partial_i f(a)$ even when both exist.*

Remark 63 (Differentiability does not imply continuity of the derivative map)**.** *Let $\Omega \subseteq V$ be open and let $f : \Omega \to W$ be a map between normed spaces. If f is Fréchet differentiable at $a \in \Omega$, then f is continuous at a. However, the existence of the derivative at a does* not *imply that the derivative map*

$$Df : \Omega \to B(V, W), \qquad x \mapsto (Df)_x,$$

is continuous at a. Thus the condition $f \in C^1$ is strictly stronger than "f is differentiable": by definition, $f \in C^1$ means that f is Fréchet differentiable at every point of Ω and that Df is continuous on Ω.

Taylor's Theorem

Standing conventions

Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed $\mathbb{R}$-vector spaces, let $\Omega \subseteq V$ be open, let $f : \Omega \to W$ be a map, and fix $a \in \Omega$. For $x \in \Omega$ we write the increment variable

$$h := x - a \in V,$$

so that $x = a + h$.

Conceptual meaning: approximation by a polynomial in the increment

The guiding idea is that, near the base point a, the values $f(a+h)$ can be approximated by an explicit polynomial in h whose coefficients are determined by the derivatives of f at a. The approximation improves as one keeps more terms, and the remaining error is of strictly higher order than the last retained term.

Zeroth order. The crudest approximation is $f(a+h) \approx f(a)$.

First order (Fréchet differentiability). If f is Fréchet differentiable at a, then there exists a bounded linear map $(Df)_a : V \to W$ such that

$$f(a+h) = f(a) + (Df)_a(h) + r_1(h),$$

with the quantitative smallness condition

$$\frac{\|r_1(h)\|_W}{\|h\|_V} \to 0 \qquad (h \to 0).$$

Conceptually: $f(a) + (Df)_a(h)$ is the correct first-order (linear) approximation of f at a, and the remainder is negligible compared to $\|h\|_V$.

Definition 34 (Taylor polynomial of degree n at a)**.** *Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed $\mathbb{R}$-vector spaces, let $\Omega \subseteq V$ be open, let $f : \Omega \to W$, and fix $a \in \Omega$. Let $n \in \mathbb{N}$ and assume that f is n-times Fréchet differentiable at a, so that for each $k \in \{1, \dots, n\}$,*

$$(D^k f)_a : V^k \to W$$

is a bounded k-linear map. The Taylor polynomial of degree n at a *is the map $T_n : V \to W$ defined by*

$$T_n(h) := f(a) + \sum_{k=1}^{n} \frac{1}{k!} (D^k f)_a(\underbrace{h, \dots, h}_{k \text{ times}}), \qquad h \in V.$$

Definition 35 (Taylor expansion of order n at a)**.** *With the hypotheses and notation of the previous definition, we say that f* admits a Taylor expansion of order n at a *if there exists a map*

$$R_n : \{h \in V : \ a + h \in \Omega\} \to W$$

such that for all h with $a + h \in \Omega$,

$$f(a+h) = T_n(h) + R_n(h),$$

and the remainder satisfies

$$\frac{\|R_n(h)\|_W}{\|h\|_V^n} \to 0 \qquad (h \to 0).$$

Remark 64 (Meaning)**.** *The map T_n is a polynomial in the increment h (a finite sum of homogeneous terms of degrees $1, \dots, n$). The remainder condition means that $R_n(h)$ is negligible compared to $\|h\|_V^n$ as $h \to 0$. Conceptually, T_n captures the behaviour of f up to order n near a, and the first neglected behaviour is of strictly higher order as $h \to 0$.*

One-variable special case (intuition)

Let $I \subseteq \mathbb{R}$ be open, let $g : I \to W$, and fix $t_0 \in I$. If g is n-times differentiable at t_0, then for small h one has the classical Taylor form

$$g(t_0 + h) = \sum_{k=0}^{n} \frac{1}{k!} g^{(k)}(t_0)\, h^k + (\text{higher order in } h).$$

Example: A concrete second-order Taylor polynomial and expansion in $\mathbb{R}^2$

Example 6 (Example: A concrete second-order Taylor polynomial and expansion in $\mathbb{R}^2$)**.** *Let $V = \mathbb{R}^2$ (with the Euclidean norm), let $W = \mathbb{R}$, and define*

$$f : \mathbb{R}^2 \to \mathbb{R}, \qquad f(x_1, x_2) = x_1^2 x_2.$$

Fix the base point $a = (a_1, a_2) \in \mathbb{R}^2$ and write increments as $h = (h_1, h_2)$, so that $a + h = (a_1 + h_1, a_2 + h_2)$.
We compute a Taylor polynomial of order 2 at a. First,

$$f(a) = a_1^2 a_2.$$

Next, the first derivative $(Df)_a : \mathbb{R}^2 \to \mathbb{R}$ is given (in coordinates) by

$$(Df)_a(h) = \partial_1 f(a)\, h_1 + \partial_2 f(a)\, h_2,$$

where

$$\partial_1 f(x_1, x_2) = 2x_1 x_2, \qquad \partial_2 f(x_1, x_2) = x_1^2.$$

Hence

$$(Df)_a(h) = (2a_1 a_2)\, h_1 + (a_1^2)\, h_2.$$

For the second derivative, note that the Hessian matrix of f at (x_1, x_2) is

$$H_f(x_1, x_2) = \begin{bmatrix} \partial_1^2 f & \partial_1 \partial_2 f \\ \partial_2 \partial_1 f & \partial_2^2 f \end{bmatrix} = \begin{bmatrix} 2x_2 & 2x_1 \\ 2x_1 & 0 \end{bmatrix}.$$

Thus at a,

$$H_f(a) = \begin{bmatrix} 2a_2 & 2a_1 \\ 2a_1 & 0 \end{bmatrix}.$$

The associated bilinear map $(D^2 f)_a : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ satisfies

$$(D^2 f)_a(h, h) = h^\top H_f(a)\, h = 2a_2 h_1^2 + 4a_1 h_1 h_2.$$

Therefore the Taylor polynomial of degree 2 at a is

$$T_2(h) = f(a) + (Df)_a(h) + \frac{1}{2}(D^2 f)_a(h, h),$$

that is,

$$T_2(h) = a_1^2 a_2 + (2a_1 a_2)\, h_1 + (a_1^2)\, h_2 + \frac{1}{2}\big(2a_2 h_1^2 + 4a_1 h_1 h_2\big),$$

so

$$T_2(h) = a_1^2 a_2 + (2a_1 a_2)\, h_1 + (a_1^2)\, h_2 + a_2 h_1^2 + 2a_1 h_1 h_2.$$

Finally, compute the remainder explicitly. Expanding $f(a+h)$ gives

$$\begin{aligned} f(a+h) &= (a_1+h_1)^2(a_2+h_2) \\ &= (a_1^2+2a_1h_1+h_1^2)(a_2+h_2) \\ &= a_1^2a_2+(2a_1a_2)h_1+(a_1^2)h_2+a_2h_1^2+2a_1h_1h_2+h_1^2h_2. \end{aligned}$$

Comparing with $T_2(h)$ yields

$$f(a+h) = T_2(h)+R_2(h), \qquad R_2(h) = h_1^2h_2.$$

Moreover, as $h \to 0$,

$$\frac{|R_2(h)|}{\|h\|_2^2} = \frac{|h_1|^2|h_2|}{\|h\|_2^2} \le |h_2| \le \|h\|_2 \longrightarrow 0,$$

so indeed $R_2(h) = o(\|h\|_2^2)$. Thus f admits a Taylor expansion of order 2 at a with Taylor polynomial T_2 and remainder R_2 as above.

Example: A concrete third-order Taylor polynomial and expansion in $\mathbb{R}^2$

Example 7 (Example: A concrete third-order Taylor polynomial and expansion in $\mathbb{R}^2$)**.** *Let $V = \mathbb{R}^2$ (with the Euclidean norm), let $W = \mathbb{R}$, and define*

$$f : \mathbb{R}^2 \to \mathbb{R}, \qquad f(x_1,x_2) = x_1^3+x_1^2x_2+x_1x_2^2+x_2^3.$$

Fix the base point $a = (a_1,a_2) \in \mathbb{R}^2$ and write increments as $h = (h_1,h_2)$, so that $a+h = (a_1+h_1, a_2+h_2)$.
Since f is a polynomial of total degree 3, its Taylor polynomial of degree 3 at a will equal $f(a+h)$ exactly (so the remainder will be 0).
Deriving the coordinate formulas for $(D^2f)_a(h,h)$ and $(D^3f)_a(h,h,h)$ from multilinearity. *Write $h = h_1e_1+h_2e_2$. By the general coordinate expansion via multilinearity (specialised to $d=2$), we have for $n=2$:*

$$\begin{aligned} (D^2f)_a(h,h) &= \sum_{i,j=1}^{2} h_ih_j\,(D^2f)_a(e_i,e_j) \\ &= h_1^2\,(D^2f)_a(e_1,e_1)+h_1h_2\,(D^2f)_a(e_1,e_2)+h_2h_1\,(D^2f)_a(e_2,e_1)+h_2^2\,(D^2f)_a(e_2,e_2). \end{aligned}$$

Using the identification $(D^2f)_a(e_i,e_j) = \partial_i\partial_jf(a)$ (with the fixed order convention) gives

$$(D^2f)_a(h,h) = \partial_1^2f(a)\,h_1^2+\partial_1\partial_2f(a)\,h_1h_2+\partial_2\partial_1f(a)\,h_2h_1+\partial_2^2f(a)\,h_2^2.$$

If, in addition, the mixed second partials agree at a (for instance under the hypotheses that ensure symmetry), then $\partial_1\partial_2f(a) = \partial_2\partial_1f(a)$ and hence

$$(D^2f)_a(h,h) = \partial_1^2f(a)\,h_1^2+2\,\partial_1\partial_2f(a)\,h_1h_2+\partial_2^2f(a)\,h_2^2.$$

Similarly, for $n=3$ the same multilinearity identity yields

$$\begin{aligned} (D^3f)_a(h,h,h) &= \sum_{i,j,k=1}^{2} h_ih_jh_k\,(D^3f)_a(e_i,e_j,e_k) \\ &= h_1^3\,(D^3f)_a(e_1,e_1,e_1) \\ &\quad + h_1^2h_2\Big[(D^3f)_a(e_1,e_1,e_2)+(D^3f)_a(e_1,e_2,e_1)+(D^3f)_a(e_2,e_1,e_1)\Big] \\ &\quad + h_1h_2^2\Big[(D^3f)_a(e_1,e_2,e_2)+(D^3f)_a(e_2,e_1,e_2)+(D^3f)_a(e_2,e_2,e_1)\Big] \\ &\quad + h_2^3\,(D^3f)_a(e_2,e_2,e_2). \end{aligned}$$

Again $(D^3 f)_a(e_i, e_j, e_k) = \partial_i \partial_j \partial_k f(a)$, *so*

$$\begin{aligned}(D^3 f)_a(h,h,h) &= \partial_1^3 f(a)\, h_1^3 + \Big(\partial_1^2\partial_2 f(a) + \partial_1\partial_2\partial_1 f(a) + \partial_2\partial_1^2 f(a)\Big)\, h_1^2 h_2 \\ &\quad + \Big(\partial_1\partial_2^2 f(a) + \partial_2\partial_1\partial_2 f(a) + \partial_2^2\partial_1 f(a)\Big)\, h_1 h_2^2 + \partial_2^3 f(a)\, h_2^3.\end{aligned}$$

If, in addition, the third derivative at a *is symmetric (for instance under the hypotheses that ensure symmetry), then the three terms in each bracket coincide and one obtains the standard grouped form*

$$(D^3 f)_a(h,h,h) = \partial_1^3 f(a)\, h_1^3 + 3\,\partial_1^2\partial_2 f(a)\, h_1^2 h_2 + 3\,\partial_1\partial_2^2 f(a)\, h_1 h_2^2 + \partial_2^3 f(a)\, h_2^3.$$

We compute T_3 *in the standard Taylor form*

$$T_3(h) = f(a) + (Df)_a(h) + \frac{1}{2}(D^2 f)_a(h,h) + \frac{1}{6}(D^3 f)_a(h,h,h).$$

First,

$$f(a) = a_1^3 + a_1^2 a_2 + a_1 a_2^2 + a_2^3.$$

For the first derivative, compute the partial derivatives:

$$\partial_1 f(x_1,x_2) = 3x_1^2 + 2x_1x_2 + x_2^2, \qquad \partial_2 f(x_1,x_2) = x_1^2 + 2x_1x_2 + 3x_2^2.$$

Hence

$$(Df)_a(h) = \big(3a_1^2 + 2a_1a_2 + a_2^2\big)h_1 + \big(a_1^2 + 2a_1a_2 + 3a_2^2\big)h_2.$$

For the second derivative, compute the second partial derivatives:

$$\partial_1^2 f(x_1,x_2) = 6x_1 + 2x_2, \qquad \partial_1\partial_2 f(x_1,x_2) = 2x_1 + 2x_2, \qquad \partial_2^2 f(x_1,x_2) = 2x_1 + 6x_2.$$

Thus, using the usual coordinate expression of $(D^2 f)_a(h,h)$ *in* $\mathbb{R}^2$,

$$\begin{aligned}(D^2 f)_a(h,h) &= \partial_1^2 f(a)\, h_1^2 + 2\,\partial_1\partial_2 f(a)\, h_1 h_2 + \partial_2^2 f(a)\, h_2^2 \\ &= \big(6a_1 + 2a_2\big)h_1^2 + 2\big(2a_1 + 2a_2\big)h_1h_2 + \big(2a_1 + 6a_2\big)h_2^2.\end{aligned}$$

For the third derivative, compute the third-order partial derivatives:

$$\partial_1^3 f = 6, \qquad \partial_1^2\partial_2 f = 2, \qquad \partial_1\partial_2^2 f = 2, \qquad \partial_2^3 f = 6,$$

(all constant functions). Therefore, again using the standard coordinate expansion for $(D^3 f)_a(h,h,h)$ *in* $\mathbb{R}^2$,

$$\begin{aligned}(D^3 f)_a(h,h,h) &= \partial_1^3 f(a)\, h_1^3 + 3\,\partial_1^2\partial_2 f(a)\, h_1^2 h_2 + 3\,\partial_1\partial_2^2 f(a)\, h_1 h_2^2 + \partial_2^3 f(a)\, h_2^3 \\ &= 6h_1^3 + 3\cdot 2\, h_1^2 h_2 + 3\cdot 2\, h_1 h_2^2 + 6h_2^3 \\ &= 6h_1^3 + 6h_1^2 h_2 + 6h_1 h_2^2 + 6h_2^3.\end{aligned}$$

Substituting into the Taylor polynomial gives

$$\begin{aligned}T_3(h) &= f(a) + (Df)_a(h) + \frac{1}{2}(D^2 f)_a(h,h) + \frac{1}{6}(D^3 f)_a(h,h,h) \\ &= f(a) + \big(3a_1^2 + 2a_1a_2 + a_2^2\big)h_1 + \big(a_1^2 + 2a_1a_2 + 3a_2^2\big)h_2 \\ &\quad + \frac{1}{2}\Big(\big(6a_1 + 2a_2\big)h_1^2 + 2\big(2a_1 + 2a_2\big)h_1h_2 + \big(2a_1 + 6a_2\big)h_2^2\Big) \\ &\quad + \frac{1}{6}\Big(6h_1^3 + 6h_1^2h_2 + 6h_1h_2^2 + 6h_2^3\Big).\end{aligned}$$

Equivalently,

$$\begin{aligned} T_3(h) = f(a) &+ \big(3a_1^2 + 2a_1a_2 + a_2^2\big)h_1 + \big(a_1^2 + 2a_1a_2 + 3a_2^2\big)h_2 \\ &+ \big(3a_1 + a_2\big)h_1^2 + \big(2a_1 + 2a_2\big)h_1h_2 + \big(a_1 + 3a_2\big)h_2^2 \\ &+ h_1^3 + h_1^2h_2 + h_1h_2^2 + h_2^3. \end{aligned}$$

Finally, since f is a polynomial of total degree 3, we have the identity

$$f(a+h) = T_3(h) \qquad \text{for all } h \in \mathbb{R}^2,$$

so the remainder in the Taylor expansion of order 3 is $R_3(h) = 0$.

Remark 65 (Derivatives of the Taylor polynomial)**.** *Let $\Omega \subseteq V$ be open, let $f : \Omega \to W$, fix $a \in \Omega$, and let $n \in \mathbb{N}$. Assume f is n-times Fréchet differentiable at a, so that $(D^k f)_a \in B_k(V, W)$ is defined for $k = 1, \dots, n$. Define the Taylor polynomial of degree n at a by*

$$T_n(h) := f(a) + \sum_{k=1}^{n} \frac{1}{k!} (D^k f)_a(\underbrace{h, \dots, h}_{k \text{ times}}), \qquad h \in V.$$

Then T_n is a polynomial map of degree at most n (as a function of the increment h), and it satisfies:

$$(D^j T_n)_0 = (D^j f)_a \qquad \text{for each } j \in \{0, \dots, n\},$$

and

$$(D^j T_n)_h = 0 \qquad \text{for all } h \in V \text{ and all } j > n.$$

Diagonal evaluation of a multilinear map is homogeneous

Proposition 41 (Homogeneity from multilinearity)**.** *Let $k \in \mathbb{N} \setminus \{0\}$ and let $T : V^k \to W$ be a k-linear map. Define*

$$P : V \to W, \qquad P(h) := T(\underbrace{h, \dots, h}_{k \text{ times}}).$$

Then P is homogeneous of degree k, meaning that for all $t \in \mathbb{R}$ and all $h \in V$,

$$P(th) = t^k\, P(h).$$

Proof. Fix $t \in \mathbb{R}$ and $h \in V$. By k-linearity of T,

$$P(th) = T(th, \dots, th) = t \cdots t\, T(h, \dots, h) = t^k\, T(h, \dots, h) = t^k\, P(h).$$

□

Remark 66 (Application to higher derivatives)**.** *If $f : \Omega \to W$ is k-times Fréchet differentiable at $a \in \Omega$, then $(D^k f)_a : V^k \to W$ is k-linear. Therefore the map*

$$h \longmapsto (D^k f)_a(\underbrace{h, \dots, h}_{k \text{ times}})$$

is homogeneous of degree k.

Homogeneous Polynomials

Standing conventions

Throughout this section, $d \in \mathbb{N}$ is fixed and we work on $\mathbb{R}^d$ with its standard coordinates $x = (x_1, \dots, x_d)$. All polynomials are real-valued.

Multi-index notation

Definition 36 (Multi-index and its order)**.** *A* multi-index *is an element* $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}^d$*. Its* order *(or* total degree*) is*

$$|\alpha| := \alpha_1 + \cdots + \alpha_d.$$

Definition 37 (Associated notation)**.** *Let* $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}^d$*. We write*

$$\alpha! := \alpha_1! \cdots \alpha_d!, \qquad x^\alpha := x_1^{\alpha_1} \cdots x_d^{\alpha_d}.$$

If $f : \mathbb{R}^d \to \mathbb{R}$ *has all iterated partial derivatives of order* $|\alpha|$ *at a point* $a \in \mathbb{R}^d$*, we write*

$$\partial^\alpha f(a)$$

for the iterated partial derivative obtained by differentiating α_i *times with respect to the* i*-th coordinate, for each* $i \in \{1, \dots, d\}$*, and then evaluating at* a*.*

Remark 67 (Conceptual meaning)**.** *The entry* α_i *records how many derivatives are taken in the* i*-th coordinate direction, and* $|\alpha|$ *is the total number of derivatives. For example, in* $d = 3$*, the multi-index* $\alpha = (2, 0, 1)$ *has* $|\alpha| = 3$ *and corresponds to "two derivatives in the first coordinate, none in the second, and one in the third".*

Example 8 (Example of the notation $\alpha!$, x^α, and $\partial^\alpha f(a)$)**.** *Let* $d = 3$ *and let* $\alpha = (2, 0, 1) \in \mathbb{N}^3$*.*
(1) The factorial of the multi-index is

$$\alpha! = \alpha_1!\alpha_2!\alpha_3! = 2!\,0!\,1! = 2.$$

(2) For $x = (x_1, x_2, x_3) \in \mathbb{R}^3$ *one has*

$$x^\alpha = x_1^{\alpha_1} x_2^{\alpha_2} x_3^{\alpha_3} = x_1^2 x_2^0 x_3^1 = x_1^2 x_3.$$

(3) Let $f : \mathbb{R}^3 \to \mathbb{R}$ *be defined by*

$$f(x_1, x_2, x_3) = x_1^2 x_3 + x_2.$$

Then, by definition, $\partial^\alpha f(a)$ *means: differentiate twice with respect to* x_1*, zero times with respect to* x_2*, once with respect to* x_3*, and then evaluate at* $a = (a_1, a_2, a_3)$*. Concretely,*

$$\partial^\alpha f(a) = \partial_1^2 \partial_3 f(a).$$

Compute:

$$\partial_3 f(x_1, x_2, x_3) = x_1^2, \qquad \partial_1(\partial_3 f)(x_1, x_2, x_3) = 2x_1, \qquad \partial_1^2(\partial_3 f)(x_1, x_2, x_3) = 2.$$

Hence

$$\partial^\alpha f(a) = \partial_1^2 \partial_3 f(a) = 2,$$

which in this example is independent of a*.*

Homogeneous polynomials and coefficient extraction

Proposition 42 (Monomial expansion of homogeneous polynomials)**.** *Let $n \in \mathbb{N}$ and let $f : \mathbb{R}^d \to \mathbb{R}$ be a homogeneous polynomial of degree n. Then there exist unique coefficients $s_\alpha \in \mathbb{R}$, indexed by the multi-indices $\alpha \in \mathbb{N}^d$ with $|\alpha| = n$, such that*

$$f(x) = \sum_{|\alpha|=n} \frac{1}{\alpha!} s_\alpha x^\alpha \qquad (x \in \mathbb{R}^d).$$

Remark 68 (Conceptual meaning)**.** *The monomials x^α with $|\alpha| = n$ form a basis of the vector space of homogeneous polynomials of degree n. The proposition says that every such f has a unique coordinate expansion with respect to this basis.*

Proposition 43 (Coefficient extraction by differentiation at 0)**.** *In the setting of the previous proposition, the coefficients are given by*

$$s_\alpha = \partial^\alpha f(0) \qquad (|\alpha| = n).$$

Example: Monomial expansion and coefficient extraction

Example 9 (Example: monomial expansion and coefficient extraction)**.** *Let $d = 2$ and $n = 3$. Define a homogeneous polynomial $f : \mathbb{R}^2 \to \mathbb{R}$ by*

$$f(x_1, x_2) = 2x_1^3 + 6x_1^2 x_2 + 3x_1 x_2^2 + 5x_2^3.$$

The multi-indices $\alpha \in \mathbb{N}^2$ with $|\alpha| = 3$ are

$$(3, 0),\ (2, 1),\ (1, 2),\ (0, 3).$$

For these,

$$\alpha! = \alpha_1!\alpha_2! \quad \textit{gives} \quad (3, 0)! = 6,\ (2, 1)! = 2,\ (1, 2)! = 2,\ (0, 3)! = 6,$$

and

$$x^{(3,0)} = x_1^3, \quad x^{(2,1)} = x_1^2 x_2, \quad x^{(1,2)} = x_1 x_2^2, \quad x^{(0,3)} = x_2^3.$$

Thus f can be written in the normalised form

$$f(x) = \sum_{|\alpha|=3} \frac{1}{\alpha!} s_\alpha x^\alpha = \frac{1}{6} s_{(3,0)} x_1^3 + \frac{1}{2} s_{(2,1)} x_1^2 x_2 + \frac{1}{2} s_{(1,2)} x_1 x_2^2 + \frac{1}{6} s_{(0,3)} x_2^3,$$

so comparing coefficients yields

$$s_{(3,0)} = 12, \qquad s_{(2,1)} = 12, \qquad s_{(1,2)} = 6, \qquad s_{(0,3)} = 30.$$

Now verify the coefficient extraction formula $s_\alpha = \partial^\alpha f(0)$.
For $\alpha = (3, 0)$,

$$\partial^{(3,0)} f(0) = \partial_1^3 f(0).$$

Since $\partial_1^3(x_1^3) = 6$ and $\partial_1^3(x_1^2 x_2) = \partial_1^3(x_1 x_2^2) = \partial_1^3(x_2^3) = 0$, we get

$$\partial_1^3 f(0) = 2 \cdot 6 = 12 = s_{(3,0)}.$$

For $\alpha = (2, 1)$,

$$\partial^{(2,1)} f(0) = \partial_1^2 \partial_2 f(0).$$

Only the term $6x_1^2x_2$ contributes: indeed

$$\partial_2(6x_1^2x_2) = 6x_1^2, \qquad \partial_1^2(6x_1^2) = 12,$$

so $\partial_1^2\partial_2 f(0) = 12 = s_{(2,1)}$.
For $\alpha = (1,2)$,

$$\partial^{(1,2)} f(0) = \partial_1\partial_2^2 f(0).$$

Only the term $3x_1x_2^2$ contributes: indeed

$$\partial_2^2(3x_1x_2^2) = 6x_1, \qquad \partial_1(6x_1) = 6,$$

so $\partial_1\partial_2^2 f(0) = 6 = s_{(1,2)}$.
For $\alpha = (0,3)$,

$$\partial^{(0,3)} f(0) = \partial_2^3 f(0).$$

Only the term $5x_2^3$ contributes, and $\partial_2^3(x_2^3) = 6$, hence

$$\partial_2^3 f(0) = 5 \cdot 6 = 30 = s_{(0,3)}.$$

This confirms $s_\alpha = \partial^\alpha f(0)$ for all multi-indices α with $|\alpha| = 3$ in this example.

Translating a homogeneous polynomial and recovering it from its k-th derivative

Remark 69 (Recovering a translated homogeneous polynomial from its k-th derivative)**.** *Let $d \in \mathbb{N}$ and let $k \in \mathbb{N} \setminus \{0\}$. Let $q : \mathbb{R}^d \to \mathbb{R}$ be a homogeneous polynomial of degree k and fix $a \in \mathbb{R}^d$. Define the translated polynomial*

$$q_a : \mathbb{R}^d \to \mathbb{R}, \qquad q_a(x) := q(x-a).$$

Since q_a is a polynomial of degree k, its k-th derivative $(D^k q_a)_b$ is a k-linear map and is independent of $b \in \mathbb{R}^d$, and all derivatives of order $> k$ vanish. Moreover, q_a is recovered from its k-th derivative by the identity

$$q_a(x) = \frac{1}{k!}\,(D^k q_a)_b(\underbrace{x-a, \dots, x-a}_{k \text{ times}}) \qquad (b, x \in \mathbb{R}^d).$$

Here the factor $1/k!$ multiplies the value of the k-linear map $(D^k q_a)_b$ evaluated on $(x-a, \dots, x-a)$; it is not *a factor in front of q_a.*
In particular, if one defines

$$f(x) := \frac{1}{k!}\,q_a(x),$$

then

$$(D^k f)_a = (D^k q_a)_a,$$

so evaluating on $(u, \dots, u)$ gives

$$(D^k f)_a(\underbrace{u, \dots, u}_{k \text{ times}}) = q(u),$$

which explains the $1/k!$ normalisation used when constructing functions with prescribed k-th order behaviour at a.

Example: Constructing f from prescribed third-order data at a

Example 10. *Fix $a = (a_1, a_2) \in \mathbb{R}^2$. We want to construct a function $f : \mathbb{R}^2 \to \mathbb{R}$ whose* third-order behaviour at a *is prescribed by*

$$(D^3 f)_a(u, u, u) = 2u_1^2 u_2 \qquad (u = (u_1, u_2) \in \mathbb{R}^2).$$

Conceptual meaning. *The map $(D^3 f)_a : (\mathbb{R}^2)^3 \to \mathbb{R}$ is trilinear, so the expression $(D^3 f)_a(u, u, u)$ is a homogeneous polynomial of degree 3 in u. Conversely, given a homogeneous polynomial $q(u)$ of degree 3, one can build a degree-3 polynomial in x (centred at a) whose third derivative at a reproduces q. This is exactly the mechanism behind the cubic term $\frac{1}{3!}(D^3 f)_a(h, h, h)$ in the Taylor polynomial.*

Construction. *Define the homogeneous polynomial $q : \mathbb{R}^2 \to \mathbb{R}$ by*

$$q(u) := 2u_1^2 u_2.$$

Translate it to be centred at a by defining

$$q_a(x) := q(x - a) = 2(x_1 - a_1)^2(x_2 - a_2) \qquad (x = (x_1, x_2) \in \mathbb{R}^2).$$

Now set

$$f(x) := \frac{1}{3!}\, q_a(x) = \frac{2}{3!}(x_1 - a_1)^2(x_2 - a_2).$$

Taylor's Theorem in coordinates

Statement

Proposition 44 (Taylor's theorem in coordinates)**.** *Let $d, m \in \mathbb{N}$, let $\Omega \subseteq \mathbb{R}^d$ be open, let $a \in \Omega$, and let $f : \Omega \to \mathbb{R}^m$. Let $n \in \mathbb{N}$ and assume that f admits a Taylor expansion of order n at a in the Fréchet sense, that is: there exists a map*

$$R_n : \{h \in \mathbb{R}^d : \ a + h \in \Omega\} \to \mathbb{R}^m$$

such that for all h with $a + h \in \Omega$,

$$f(a + h) = f(a) + \sum_{k=1}^{n} \frac{1}{k!}(D^k f)_a(\underbrace{h, \ldots, h}_{k \text{ times}}) + R_n(h),$$

and

$$\frac{\|R_n(h)\|_2}{\|h\|_2^n} \to 0 \qquad (h \to 0).$$

Then one has the coordinate form

$$f(a + h) = \sum_{|\alpha| \le n} \frac{1}{\alpha!}\, \partial^\alpha f(a)\, h^\alpha + R_n(h),$$

where $h^\alpha = h_1^{\alpha_1} \cdots h_d^{\alpha_d}$ and $\partial^\alpha f(a)$ denotes the iterated partial derivative of multi-index α at a (taken componentwise when $m > 1$). Moreover,

$$\frac{\|R_n(h)\|_2}{\|h\|_2^n} \to 0 \qquad (h \to 0).$$

Taylor's Theorem with Lagrange Remainder

Standing conventions

Throughout this section, $I \subseteq \mathbb{R}$ is an open interval and $g : I \to \mathbb{R}$ is a function. Fix $t_0 \in I$ and let $n \in \mathbb{N}$.

Statement

Proposition 45 (Taylor's theorem with Lagrange remainder)**.** *Let $t \in I$ and assume that g is $(n+1)$-times differentiable on an interval containing t_0 and t. Then there exists a point ξ between t_0 and t such that*

$$g(t) = \sum_{k=0}^{n} \frac{1}{k!}\, g^{(k)}(t_0)\,(t-t_0)^k + \frac{1}{(n+1)!}\, g^{(n+1)}(\xi)\,(t-t_0)^{n+1}.$$

The last term is called the Lagrange remainder*.*

Conceptual meaning

Remark 70 (What the theorem is saying)**.** *The Taylor polynomial of degree n at t_0 approximates $g(t)$ up to order n. Taylor's theorem with Lagrange remainder strengthens this by identifying the error exactly as a single $(n+1)$-st order term:*

$$g(t) - \sum_{k=0}^{n} \frac{1}{k!}\, g^{(k)}(t_0)\,(t-t_0)^k = \frac{1}{(n+1)!}\, g^{(n+1)}(\xi)\,(t-t_0)^{n+1}$$

for some intermediate point ξ between t_0 and t. Thus the error is always proportional to $(t-t_0)^{n+1}$, and the proportionality constant is an $(n+1)$-st derivative evaluated at some point in between. The point ξ is not given explicitly; the theorem asserts its existence.

Remark 71 (Practical error bound)**.** *If there exists $M \geq 0$ such that*

$$|g^{(n+1)}(s)| \leq M$$

for all s between t_0 and t, then the Lagrange remainder immediately yields the estimate

$$\left| g(t) - \sum_{k=0}^{n} \frac{1}{k!}\, g^{(k)}(t_0)\,(t-t_0)^k \right| \leq \frac{M}{(n+1)!}\,|t-t_0|^{n+1}.$$

Remark 72 (Relation between the two notations for the Taylor polynomial)**.** *In one variable, the Taylor polynomial of degree n at t_0 is naturally written as a polynomial in the increment h:*

$$T_n(h) := \sum_{k=0}^{n} \frac{1}{k!}\, g^{(k)}(t_0)\, h^k.$$

When one evaluates at a point $t \in I$, the increment is $h = t - t_0$, so the same polynomial appears as

$$T_n(t-t_0) = \sum_{k=0}^{n} \frac{1}{k!}\, g^{(k)}(t_0)\,(t-t_0)^k.$$

Thus the expression $\sum_{k=0}^{n} \frac{1}{k!}\, g^{(k)}(t_0)(t-t_0)^k$ is simply the Taylor polynomial evaluated at $h = t - t_0$.

Remark 73 (Why "three terms" appear, and how this relates to the Lagrange remainder)**.** *In the Fréchet-form Taylor expansion*

$$f(a+h) = f(a) + \sum_{k=1}^{n} \frac{1}{k!}(D^k f)_a(\underbrace{h, \dots, h}_{k \text{ times}}) + R_n(h),$$

the right-hand side is split into three parts: the constant term $f(a)$ (order 0), the Taylor polynomial terms of orders 1 through n, and the remainder $R_n(h)$.
In one variable, the Taylor polynomial is often written with the index starting at $k = 0$:

$$\sum_{k=0}^{n} \frac{1}{k!}\, g^{(k)}(t_0)\,(t - t_0)^k,$$

and the $k = 0$ term is exactly $g(t_0)$. Thus the apparent difference is only notational: the constant term is either written separately (as $f(a)$) or included as the $k = 0$ term of the sum.

Taylor Approximations of Standard Functions

Standing conventions

Throughout this section, $n \in \mathbb{N}$ and $x \in \mathbb{R}$ is assumed to be sufficiently close to 0 so that all expressions are defined (in particular $\ln(1+x)$ is considered for $x > -1$).

$O(|x|^N)$ notation

Definition 38 (Big-O at 0)**.** *Let $N \in \mathbb{N}$ and let $f, g : D \to \mathbb{R}$ be functions with 0 an accumulation point of $D \subseteq \mathbb{R}$. We write*

$$f(x) = g(x) + O(|x|^N) \qquad (x \to 0,\ x \in D)$$

if there exist constants $C \geq 0$ and $\delta > 0$ such that for all $x \in D$,

$$|x| < \delta \quad \Longrightarrow \quad |f(x) - g(x)| \leq C|x|^N.$$

Standard Taylor approximations at 0

Proposition 46 (Taylor approximations of standard functions)**.** *For every $n \in \mathbb{N}$, as $x \to 0$ one has:*

$$\exp(x) = \sum_{k=0}^{n} \frac{x^k}{k!} + O(|x|^{n+1}),$$

$$\sin(x) = \sum_{k=0}^{n} (-1)^k \frac{x^{2k+1}}{(2k+1)!} + O(|x|^{2n+3}),$$

$$\cos(x) = \sum_{k=0}^{n} (-1)^k \frac{x^{2k}}{(2k)!} + O(|x|^{2n+2}),$$

and, for $x \to 0$ with $x > -1$,

$$\ln(1+x) = \sum_{k=1}^{n} (-1)^{k+1} \frac{x^k}{k} + O(|x|^{n+1}).$$